

IIT Questions

One–Option–Correct Questions

- Q 1. For all $x \in (0, 1)$,
 (a) $e^x < 1 + x$ (b) $\log_e(1 + x) < x$ (c) $\sin x > x$ (d) $\log_e x > x$
- Q 2. The incentre of the triangle with vertices $(1, \sqrt{3})$, $(0, 0)$ and $(2, 0)$ is
 (a) $\left(1, \frac{\sqrt{3}}{2}\right)$ (b) $\left(\frac{2}{3}, \frac{1}{\sqrt{3}}\right)$ (c) $\left(\frac{2}{3}, \frac{\sqrt{3}}{2}\right)$ (d) $\left(1, \frac{1}{\sqrt{3}}\right)$
- Q 3. The line $x - 1 = 0$ is the directrix of the parabola $y^2 - kx + 8 = 0$. Then one
 (a) g (b) 8 (c) 4 (d)
- Q 4. The domain of definition of the function $y(x)$ given by $2^x + 2^y = 2$ is
 (a) $(0, 1]$ (b) $[0, 1]$ (c) $(-\infty, 0]$ (d) $(-\infty, 1)$
- Q 5. The triangle PQR is inscribed in the circle $x^2 + y^2 = 25$. If Q and R have coordinates $(3, 4)$ and $(-4, 3)$, respectively then $\angle QPR$ is equal to
 (a) $\frac{\pi}{2}$ (b) $\frac{\pi}{3}$ (c) $\frac{\pi}{4}$ (d) $\frac{\pi}{6}$
- Q 6. Let $f(x) = \int e^x (x - 1)(x - 2) dx$. Then f decreases in the interval
 (a) $(-\infty, -2)$ (b) $(-2, -1)$ (c) $(1, 2)$ (d) $(2, +\infty)$
- Q 7. Let $g(x) = \int_0^x f(t) dt$, where $\frac{1}{2} \leq f(t) \leq 1$ for $t \in [0, 1]$ and $0 \leq f(t) \leq \frac{1}{2}$ for $t \in (1, 2]$. Then
 (a) $-\frac{3}{2} \leq g(2) < \frac{1}{2}$ (b) $0 \leq g(2) < 2$ (c) $\frac{3}{2} < g(2) \leq \frac{5}{2}$ (d) $2 < g(2) < 4$
- Q 8. If $x^2 + y^2 = 1$ then $\left(\text{where } y', \frac{dy}{dx}, y'' = \frac{d^2y}{dx^2} \right)$
 (a) $yy'' - 2y'^2 + 1 = 0$ (b) $yy'' + y'^2 + 1 = 0$ (c) $yy'' - y'^2 - 1 = 0$ (d) $yy'' + 2y'^2 + 1 = 0$
- Q 9. If the vectors a , b and c are represented by the sides BC , CA and AB , respectively, of the $\triangle ABC$ then
 (a) $\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a} = 0$ (b) $\vec{a} \times \vec{b} = \vec{b} \times \vec{c} = \vec{c} \times \vec{a}$ (c) $\vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{c} = \vec{c} \cdot \vec{a}$ (d) $\vec{a} \times \vec{b} + \vec{b} \times \vec{c} + \vec{c} \times \vec{a} = \vec{0}$
- Q 10. A pole stands vertically inside a triangular park ABC . If the angle of elevation of the top of the pole from each corner of the park is the same then in the $\triangle ABC$, the foot of the pole is at the
 (a) centroid (b) circumcentre (c) incentre (d) orthocentre
- Q 11. Let $f(\theta) = (\sin \theta)(\sin \theta + \sin 3\theta)$. Then
 (a) $f(\theta) \geq 0$ only when $\theta \geq 0$ (b) $f(\theta) \leq 0$ for all real θ
 (c) $f(\theta) \geq 0$ for all real θ (d) $f(\theta) \leq 0$ only when $\theta \leq 0$
- Q 12. If $x + y = k$ is a normal to the parabola $y^2 = 12x$ then k is
 (a) 3 (b) 9 (c) -9 (d) -3
- Q 13. If z_1, z_2, z_3 are complex numbers such that $|z_1| = |z_2| = |z_3| = 1$ then $|z_1 + z_2 + z_3|$ is
 (a) equal to 1 (b) less than 1 (c) greater than 3 (d) equal to 3
- Q 14. If $\arg z < 0$ then $\arg(-z) - \arg z$ is equal to
 (a) π (b) $-\pi$ (c) $-\frac{\pi}{2}$ (d) $\frac{\pi}{2}$
- Q 15. If a, b, c, d are positive real numbers such that $a + b + c + d = 2$ then $M = (a + b)(c + d)$ satisfies the relation
 (a) $0 < M \leq 1$ (b) $1 \leq M \leq 2$ (c) $2 \leq M \leq 3$ (d) $3 \leq M \leq 4$
- Q 16. If the circles $x^2 + y^2 + 2x + 2ky + 6 = 0$ and $x^2 + y^2 + 2ky + k = 0$ intersect orthogonally then k is
 (a) 2 or $-\frac{3}{2}$ (b) -2 or $-\frac{3}{2}$ (c) 2 or $-\frac{3}{2}$ (d) -2 or $\frac{3}{2}$

- Q 17. Let PS be the median of the triangle with vertices P(2, 2), Q(6, -1) and R(7, 3). The equation of the line passing through (1, -1) and parallel to PS is
 (a) $2x - 9y - 7 = 0$ (b) $2x - 9y - 11 = 0$ (c) $2x + 9y - 11 = 0$ (d) $2x + 9y + 7 = 0$
- Q 18. If $\vec{a}, \vec{b}$ and $\vec{c}$ are unit coplanar vectors then the scalar triple product $[\vec{2a} - \vec{b}, \vec{2b} - \vec{c}, \vec{2c} - \vec{a}]$ is equal to
 (a) 0 (b) 1 (c) $-\sqrt{3}$ (d) $-\sqrt{3}$
- Q 19. Let the vectors $\vec{a}, \vec{b}, \vec{c}$ and $\vec{d}$ be such that $(\vec{a} \times \vec{b}) \times (\vec{c} \times \vec{d}) = \vec{0}$. Let P_1 and P_2 be planes determined by pairs of vectors a, b and c, d , respectively. Then the angle between P_1 and P_2 is
 (a) 0 (b) $\frac{\pi}{4}$ (c) $\frac{\pi}{3}$ (d) $\frac{\pi}{2}$
- Q 20. In a $\triangle ABC$, $2ac \cdot \sin \frac{A-B+C}{2}$ is equal to
 (a) $a^2 + b^2 - c^2$ (b) $c^2 + a^2 - b^2$ (c) $b^2 - c^2 - a^2$ (d) $c^2 - a^2 - b^2$
- Q 21. In a $\triangle ABC$, let $\angle C = \frac{\pi}{2}$. If r and R are the inradius and the circumradius, respectively, of the triangle then $2(r + R)$ is equal to
 (a) $a + b$ (b) $b + c$ (c) $c + a$ (d) $a + b + c$
- Q 22. If $f(x) = \begin{cases} e^{\cos x} \cdot \sin x & \text{for } |x| \leq 2 \\ 2 & \text{otherwise} \end{cases}$ then $\int_{-2}^3 f(x) dx$ is equal to
 (a) 0 (b) 1 (c) 2 (d) 3
- Q 23. The value of $\int_{e^{-1}}^{e^2} \left| \frac{\log_e x}{x} \right| dx$ is
 (a) $\frac{3}{2}$ (b) $\frac{5}{2}$ (c) 3 (d) 5
- Q 24. If the system of equations $x - ky - z = 0$, $kx - y - z = 0$ and $x + y - z = 0$ has a nonzero solution then the possible values of k are
 (a) -1, 2 (b) 1, 2 (c) 0, 1 (d) -1, 1
- Q 25. Consider an infinite geometric series with the first* term a and the common ratio r . If its sum is 4 and the second term is - then (a, r) has the value
 (a) $\left(\frac{7}{4}, \frac{3}{7}\right)$ (b) $\left(2, \frac{3}{8}\right)$ (c) $\left(\frac{3}{2}, \frac{1}{2}\right)$ (d) $\left(3, \frac{1}{4}\right)$
- Q 26. For $x \in \mathbb{R}$, $\lim_{x \rightarrow \infty} \left(\frac{x-3}{x+2}\right)^x$ is equal to
 (a) e (b) e^{-1} (c) e^{-5} (d) e^5
- Q 27. If the normal to the curve $y = f(x)$ at the point (3, 4) makes an angle $\frac{3\pi}{4}$ with the positive x-axis then $f'(3)$ is equal to
 (a) -1 (b) $-\frac{3}{4}$ (c) $\frac{4}{3}$ (d) 1
- Q 28. For the equation $3x^2 + px + 3 = 0$ ($p > 0$), if one of the roots is square of the other then p is equal to
 (a) $\frac{1}{3}$ (b) 1 (c) 3 (d) $\frac{2}{3}$
- Q 29. If α and β ($\alpha < \beta$) are the roots of the equation $x^2 + bx + c = 0$, where $c < 0 < b$ then
 (a) $0 < \alpha < \beta$ (b) $\alpha < 0 < \beta > |\alpha|$ (c) $\alpha < \beta < 0$ (d) $\alpha < 0 < |\alpha| < \beta$
- Q 30. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a function. Define $g: \mathbb{R} \rightarrow \mathbb{R}$ by $g(x) = |f(x)|$ for all x . Then g is
 (a) onto if f is onto (b) one-to-one if f is one-to-one
 (c) continuous if f is continuous (d) differentiable if f is differentiable

Q 31. For $2 \leq r \leq n$, $\binom{n}{r} + 2\binom{n}{r-1} + \binom{n}{r-2}$ is equal to

- (a) $\binom{n+1}{r-1}$ (b) $2\binom{n+1}{r+1}$ (c) $2\binom{n+2}{r}$ (d) $\binom{n+2}{r}$

Q 32. If $b > a$ then the equation $(x-a)(x-b) = 1$ has

- (a) both roots in $[a, b]$ (b) both roots in $(-\infty, a)$
(c) both roots in $(b, +\infty)$ (d) one root in $(-\infty, a)$ and the other in $(b, +\infty)$

Q 33. Let

$$f(x) = \begin{cases} |x| & \text{for } 0 < |x| \leq 2 \\ 1 & \text{for } x = 0. \end{cases}$$

Then at $x = 0$, f has

- (a) a local maximum (b) no local maximum (c) a local minimum (d) no extremum

Q 34. How many different nine-digit numbers can be formed from the digits of the number 223355888 by rearrangement of the digits so that the odd digits occupy even places?

- (a) 16 (b) 36 (c) 60 (d) 180

Q 35. Consider the following statements S and R.

S: Both $\sin x$ and $\cos x$ are decreasing functions in $\left(\frac{\pi}{2}, \pi\right)$.

R: If a differentiable function decreases in (a, b) then its derivative also decreases in (a, b) . Which of the following is true?

- (a) Both S and R are wrong.
(b) Both S and R are correct but R is not the correct explanation for S.
(c) S is correct and R is the correct explanation for S.
(d) S is correct and R is wrong.

Q 36. For $x > 0$ $\lim_{x \rightarrow 0} \left\{ (\sin x)^{1/x} + \left(\frac{1}{x}\right)^{\sin x} \right\}$ is

- (a) 0 (b) -1 (c) 1 (d) 2

Q 37. Given an isosceles triangle whose inradius is $\sqrt{3}$. If one of the angles of the triangle is 120° then the area of the triangle in square units is

- (a) 4π (b) $12 + 7\sqrt{3}$ (c) $12 - 7\sqrt{3}$ (d) $7 + 12\sqrt{3}$

Q 38. Let $\vec{a} = \vec{i} + 2\vec{j} + \vec{k}$, $\vec{b} = \vec{i} - \vec{j} + \vec{k}$ and $\vec{c} = \vec{i} + \vec{j} - \vec{k}$. A vector in the plane of $\vec{a}$ and $\vec{b}$ whose projection on $\vec{c}$ is $\frac{1}{\sqrt{3}}$ is

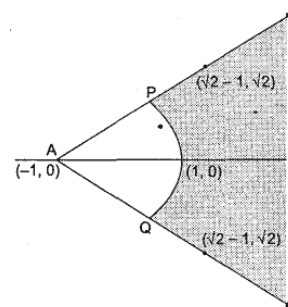
- (a) $3\vec{i} + \vec{j} - 3\vec{k}$ (b) $4\vec{i} - \vec{j} + 4\vec{k}$ (c) $4\vec{i} + \vec{j} - 4\vec{k}$ (d) $2\vec{i} + \vec{j} - 2\vec{k}$

Q 39. A plane passes through $(1, -2, 1)$ and is perpendicular to the planes $2x - 2y + 2z = 0$ and $x - y + 2z = 4$. Then the distance of that plane from the point $(1, 2, 2)$ is

- (a) 0 (b) 1 (c) $\sqrt{2}$ (d) $2\sqrt{2}$

Q 40. The locus of Z representing the complex number z and lying in the shaded region is best represented by

- (a) $z: |z+1| > 2, |\arg(z+1)| < \frac{\pi}{4}$
(b) $z: |z-1| > 2, |\arg(z-1)| < \frac{\pi}{4}$
(c) $z: |z+1| < 2, |\arg(z+1)| < \frac{\pi}{2}$
(d) $z: |z-1| < 2, |\arg(z-1)| < \frac{\pi}{2}$



One-or-More-Options-Correct Questions

- Q 41. If $f(x) = \min\{1, x^2, x^3\}$ then
 (a) $f(x)$ is continuous $\forall x \in \mathbb{R}$
 (b) $f'(x) > 0 \forall x > 1$
 (c) $f(x)$ is not differentiable but continuous $\forall x \in \mathbb{R}$
 (d) $f(x)$ is not differentiable for two values of x
- Q 42. If a hyperbola passes through the focus of the ellipse $\frac{x^2}{25} + \frac{y^2}{16} = 1$ and its transverse and conjugate axes coincide with the major and minor axes of the ellipse, the product of their eccentricities being 1 then
 (a) the equation of the hyperbola is $\frac{x^2}{9} - \frac{y^2}{16} = 1$
 (b) the equation of the hyperbola is $\frac{x^2}{9} - \frac{y^2}{25} = 1$
 (c) the focus of the hyperbola is $(5, 0)$
 (d) the focus of the hyperbola is $(5\sqrt{3}, 0)$

Comprehension-Type Questions

- Q 43. Let ABCD be a square of sides of length 2 units. C_1 is the circle touching all the sides of the square and C_2 is the circle circumscribing the square. l is a line through A.
 (i) If P is a point on C_1 and Q is a point on C_2 then

$$\frac{PA^2 + PB^2 + PC^2 + PD^2}{QA^2 + QB^2 + QC^2 + QD^2}$$
 equals
 (a) $\frac{3}{4}$ (b) $\frac{5}{4}$ (c) 1 (d) $\frac{1}{2}$
 (ii) A circle touches the line l and the circle C_1 externally such that both the circles are on the same side of the line. The locus of the centre of the circle is
 (a) part of a straight line (b) parabola (c) ellipse (d) hyperbola
 (iii) A line m through A is drawn parallel to BD. A point S moves such that its distances from BD and the vertex A are equal. If the locus of S cuts the line m at T_2 and T_3 , and AC at T_1 then $(T_1T_2T_3)$ in square units is
 (a) $\frac{1}{2}$ (b) $\frac{2}{3}$ (c) 1 (d) 2

- Q 44. Let $A = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 3 & 2 & 1 \end{bmatrix}$ and U_1, U_2, U_3 are three column matrices such that

$$AU_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, AU_2 = \begin{bmatrix} 2 \\ 3 \\ 0 \end{bmatrix} \text{ and } AU_3 = \begin{bmatrix} 2 \\ 3 \\ 1 \end{bmatrix}.$$

U is a 3×3 matrix whose first column is the column in U_1 , the second column is the column in U_2 and the third column is the column in U_3 . On the basis of this, answer the following questions.

- (i) The value of $|U|$ is
 (a) 3 (b) -3 (c) $\frac{3}{2}$ (d) 2
- (ii) The sum of the elements of U^{-1} is
 (a) 0 (b) -1 (c) 1 (d) 3
- (iii) The sum of the elements of the product $\begin{bmatrix} 3 & 2 & 0 \end{bmatrix} U \begin{bmatrix} 2 \\ 3 \\ 0 \end{bmatrix}$ is

(a) $\frac{5}{2}$

(b) $\frac{3}{2}$

(c) 4

(d) 5

Matching Questions**More than one may match with the same.**

- Q 45. (i) $\int_0^{\pi/2} (\sin x)^{\cos x} \cdot [\cos x \cot x - \log(\sin x)^{\sin x}] dx$ (a) 1
 (ii) The area bounded by $-4y^2 = x$ and $x - 1 = -5y^2$ is (b) 0
 (iii) Cosine of the angle of intersection of curves $y = 3^{x-1} \log x$ and $y = x^x - 1$ is (c) $61 \log 2$
 (iv) $\lim_{x \rightarrow \infty} \frac{\left(\int_0^x e^{x^2} dx \right)^2}{\int_0^x e^{2x^2} dx}$ is equal to (d) $\frac{4}{3}$

Complete the following statements.

- Q 46. If the roots of the equation $x^2 - 10cx - 11d = 0$ are a, b and those of $x^2 - 10ax - 11b = 0$ are c, d , then the positive value of $a + b + c + d$ is (a, b, c and d being distinct numbers)
- Q 47. If $a_n = \frac{3}{4} - \left(\frac{3}{4}\right)^2 + \left(\frac{3}{4}\right)^3 - 000 + (-1)^{n-1} \cdot \left(\frac{3}{4}\right)^n$ and $b_n = 1 - a_n$ then the smallest natural number n_0 , such that $b_n > a_n \forall n > n_0$, is

Answers

- | | | | | | | | | | |
|------|-------|--------|---------|--------|-----|-----|-----|-----|-----|
| 1b | 2d | 3c | 4d | 5c | 6c | 7b | 8b | 9b | 10b |
| 11c | 12c | 13a | 14a | 15a | 16a | 17d | 18a | 19a | 20b |
| 21a | 22c | 23b | 24f | 25d | 26c | 27d | 28c | 29b | 30c |
| 31d | 32d | 33a | 34c | 35d | 36c | 37b | 38b | 39d | 40a |
| 41ac | 42ac | | | | | | | | |
| 43. | (i) a | (ii) b | (iii) c | | | | | | |
| 44. | (i) a | (ii) a | (iii) d | | | | | | |
| 45. | (i) a | (i) d | (iii) a | (iv) b | | | | | |
| 46. | 1210 | | | | | | | | |
| 47. | 5 | | | | | | | | |

IIT Questions

One–Option–Correct Questions

- Q 1. Let α, β be roots of $x^2 - x + p = 0$ and γ, δ be the roots of $x^2 - 4x + q = 0$. If $\alpha, \beta, \gamma, \delta$ are in GP then the integral values of p and q , respectively, are
 (a) $-2, -32$ (b) $-2, 3$ (c) $-6, 3$ (d) $-6, -32$
- Q 2. In the binomial expansion of $(a - b)^n$ ($n \geq 5$), the sum of the 5th and 6th terms is zero. Then $\frac{a}{b}$ is equal to
 (a) $\frac{1}{6}(n-5)$ (b) $\frac{1}{5}(n-4)$ (c) $\frac{5}{n-4}$ (d) $\frac{6}{n-5}$
- Q 3. Let $f(x) = (1 + b^2)x^2 + 2bx + 1$ and $m(b)$ the minimum value of $f(x)$ for a given b . As b varies, the range of $m(b)$ is
 (a) $[0, 1]$ (b) $\left(0, \frac{1}{2}\right]$ (c) $\left[\frac{1}{2}, 1\right]$ (d) $(0, 1)$
- Q 4. The number of distinct real roots of

$$\begin{vmatrix} \sin x & \cos x & \cos x \\ \cos x & \sin x & \cos x \\ \cos x & \cos x & \sin x \end{vmatrix} = 0$$

 in the interval $-\frac{\pi}{4} \leq x \leq \frac{\pi}{4}$ is
 (a) 0 (b) 2 (c) 1 (d) 3
- Q 5. Let $E = \{1, 2, 3, 4\}$ and $F = \{1, 2\}$. Then the number of onto functions from E to F is
 (a) 14 (b) 16 (c) 12 (d) 8
- Q 6. Let T_n denote the number of triangles which can be formed using the vertices of a regular polygon on n sides. If $T_{n+1} - T_n = 21$ then n equals
 (a) 5 (b) 7 (c) 6 (d) 4
- Q 7. The complex numbers z_1, z_2 and z_3 satisfying $\frac{z_1 - z_3}{z_2 - z_3} = \frac{1 - \sqrt{3}i}{2}$ are the vertices of a triangle which is
 (a) of area = 0 (b) right-angled isosceles (c) equilateral (d) obtuse-angled isosceles
- Q 8. If the sum of the first $2n$ terms of $2, 5, 8, \dots$ is equal to the sum of the first n terms of $57, 59, 61, \dots$ then n is equal to
 (a) 10 (b) 12 (c) 11 (d) 13
- Q 9. Let z_1 and z_2 be n th roots of unity which are ends of a line segment that subtend a right angle at the origin. Then n must be of the form
 (a) $4k + 1$ (b) $4k + 2$ (c) $4k + 3$ (d) $4k$
- Q 10. Let the positive numbers a, b, c, d be in AP. Then abc, abd, acd, bcd are
 (a) not in AP/GP/HP (b) in AP (c) in GP (d) in HP
- Q 11. Let $f(x) = \frac{\alpha x}{x+1}$, where $x \neq -1$. Then for what value of α is $f\{f(x)\} = x$ true?
 (a) $\sqrt{2}$ (b) $-\sqrt{2}$ (c) 1 (d) -1
- Q 12. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a function defined by $f(x) = \max\{x, x^3\}$. The set of all points where $f(x)$ is not differentiable is
 (a) $\{-1, 1\}$ (b) $\{-1, 0\}$ (c) $\{0, 1\}$ (d) $\{-1, 0, 1\}$
- Q 13. Let $f: (0, +\infty) \rightarrow \mathbb{R}$ and $F(x) = \int_0^x f(t) dx$. If $F(x^2) = x^2(1 + x)$ then $f(4)$ equals
 (a) $\frac{5}{4}$ (b) 7 (c) 4 (d) 2

- Q 14. The left-hand derivative of $f(x) = [x]\sin nx$ at $x = kc$, where k is an integer and $[x] =$ greatest integer not greater than x , is
 (a) $(-1)^k(k-1)\pi$ (b) $(-1)^{k-1} \cdot (k-1)\pi$ (c) $(-1)^k k\pi$ (d) $(-1)^{k-1} k\pi$
- Q 15. If $f(x) = xe^{x(1-x)}$ then $f(x)$ is
 (a) increasing on $\left[-\frac{1}{2}, 1\right]$ (b) decreasing on $\mathbb{R}$
 (c) increasing on $\mathbb{R}$ (d) decreasing on $\left[-\frac{1}{2}, 1\right]$
- Q 16. $\lim_{x \rightarrow 0} \frac{\sin(\pi \cos^2 x)}{x^2}$ equals
 (a) $-\pi$ (b) π (c) $\frac{\pi}{2}$ (d) 1
- Q 17. The triangle formed by the tangent to the curve $f(x) = x^2 + bx - b$ at the point $(1, 1)$ and the coordinate axes, lies in the first quadrant. If its area is 2 then the value of b is
 (a) -1 (b) 3 (c) -3 (d) 1
- Q 18. Let $g(x) = 1 + x - [x]$ and

$$f(x) = \begin{cases} -1 & \text{for } x < 0 \\ 0 & \text{for } x = 0 \\ 1 & \text{for } x > 0 \end{cases}$$

 Then for all x , $f\{g(x)\}$ is equal to
 (a) x (b) 1 (c) $f(x)$ (d) $g(x)$
- Q 19. If $f: [1, +\infty) \rightarrow [2, +\infty)$ is given by $f(x) = x + \frac{1}{x}$ then f^{-1} equals
 (a) $\frac{x + \sqrt{x^2 - 4}}{2}$ (b) $\frac{x}{1 + x^2}$ (c) $\frac{x - \sqrt{x^2 - 4}}{2}$ (d) $1 + \sqrt{x^2 - 4}$
- Q 20. The domain of $f(x) = \frac{\log_2(x+3)}{x^2 + 3x + 2}$ is
 (a) $\mathbb{R} - \{-1, -2\}$ (b) $(-2, +\infty)$ (c) $\mathbb{R} - \{-1, -2, -3\}$ (d) $(-3, +\infty) - \{-1, -2\}$
- Q 21. The value of $\int_{-\pi}^{\pi} \frac{\cos^2 x}{1 + a^x} dx$ ($a > 0$) is
 (a) π (b) $a\pi$ (c) $\frac{\pi}{2}$ (d) 2π
- Q 22. The equation of the common tangent touching the circle $(x-3)^2 + y^2 = 9$ and the parabola $y^2 = 4x$ above the x -axis, is
 (a) $\sqrt{3}y = 3x + 1$ (b) $\sqrt{3}y = -(x + 3)$ (c) $\sqrt{3}y = x + 3$ (d) $\sqrt{3}y = -(3x + 1)$
- Q 23. Let AB be a chord of the circle $x^2 + y^2 = r^2$ subtending a right angle at the centre. Then the locus of the centroid of the $\triangle PAB$ as P moves on the circle is
 (a) a parabola (b) a circle (c) an ellipse (d) a pair of straight lines
- Q 24. The number of integral values of m , for which the x -coordinate of the point of intersection of the lines $3x + 4y = 9$ and $y = mx + 1$ is also an integer, is
 (a) 2 (b) 0 (c) 4 (d) 1
- Q 25. The equation of the directrix of the parabola $y^2 + 4y + 4x + 2 = 0$ is
 (a) $x = -1$ (b) $x = 1$ (c) $x = -\frac{3}{2}$ (d) $x = \frac{3}{2}$
- Q 26. If $\vec{a}, \vec{b}$ and $\vec{c}$ are unit vectors then does not exceed $|\vec{a} - \vec{b}|^2 + |\vec{b} - \vec{c}|^2 + |\vec{c} - \vec{a}|^2$ does not exceed
 (a) 4 (b) 9 (c) 8 (d) 6
- Q 27. Which of the following functions is differentiable at $x = 0$?
 (a) $\cos |x| + |x|$ (b) $\cos |x| - |x|$ (c) $\sin |x| + |x|$ (d) $\sin |x| - |x|$
- Q 28. The number of solutions of $\log_4(x-1) = \log_2(x-3)$ is
 (a) 3 (b) 1 (c) 2 (d) 0

- Q 29. Let $\vec{a} = \vec{i} - \vec{k}$, $\vec{b} = x\vec{i} + \vec{j} + (1-x)\vec{k}$ and $\vec{c} = y\vec{i} + x\vec{j} + (1+x-y)\vec{k}$. Then $[\vec{a} \vec{b} \vec{c}]$ depends on
 (a) only x (b) only y (c) neither x nor y (d) both x and y
- Q 30. The area of the parallelogram formed by the lines $y = mx$, $y = mx + 1$, $y = nx$ and $y = nx + 1$ equals
 (a) $\frac{|m+n|}{(m-n)^2}$ (b) $\frac{2}{|m+n|}$ (c) $\frac{1}{|m+n|}$ (d) $\frac{1}{|m-n|}$
- Q 31. Let PQ and RS be tangents at the extremities of the diameter PR of a circle of radius r . If PS and RQ intersect at a point X on the circumference of the circle then $2r$ equals
 (a) $\sqrt{PQ \cdot RS}$ (b) $\frac{PQ+RS}{2}$ (c) $\frac{2PQ \cdot RS}{PQ+RS}$ (d) $\sqrt{\frac{PQ^2 + RS^2}{2}}$
- Q 32. A man from the top of a 100-metre-high tower sees a car moving towards the tower at an angle of depression of 30° . After some time, the angle of depression becomes 60° . The distance (in metres) travelled by the car during the time is
 (a) $100\sqrt{3}$ (b) $\frac{200\sqrt{3}}{3}$ (c) $\frac{100\sqrt{3}}{3}$ (d) $200\sqrt{3}$
- Q 33. If $\alpha + \beta = \frac{\pi}{2}$ and $\beta + \gamma = \alpha$ then $\tan \alpha$ equals
 (a) $2(\tan \beta + \tan \gamma)$ (b) $\tan \beta + \tan \gamma$ (c) $\tan \beta + 2\tan \gamma$ (d) $2\tan \beta + \tan \gamma$
- Q 34. If $\sin^{-1}$ for $0 < |x| < \sqrt{2}$ then x equals
 (a) $\frac{1}{2}$ (b) 1 (c) $-\frac{1}{2}$ (d) -1
- Q 35. The maximum value of $\cos \alpha_1 \cdot \cos \alpha_2 \cdot \dots \cdot \cos \alpha_n$, under the restrictions $\alpha \leq \alpha_1, \alpha_2, \dots, \alpha_n \leq \frac{\pi}{2}$ and $\cot \alpha_1 \cdot \cot \alpha_2 \cdot \dots \cdot \cot \alpha_n = 1$, is
 (a) $\frac{1}{2^{n/2}}$ (b) $\frac{1}{2^n}$ (c) $\frac{1}{2n}$ (d) 1
- Q 36. $\int \frac{x^2 - 1}{x^3 \cdot \sqrt{2x^4 - 2x^2 + 1}} dx$ is equal to
 (a) $\frac{1}{x^2} \cdot \sqrt{2x^4 - 2x^2 + 1} + c$ (b) $\frac{1}{x^3} \cdot \sqrt{2x^4 - 2x^2 + 1} + c$ (c) $\frac{1}{x} \cdot \sqrt{2x^4 - 2x^2 + 1} + c$ (d) $\frac{1}{2x^2} \cdot \sqrt{2x^4 - 2x^2 + 1} + c$
- Q 37. Let a, b, c be the lengths of sides of a scalene triangle. If the roots of the equation $x^2 + 2(a+b+c)x + 3\lambda(ab+bc+ca) = 0$ ($X \in \mathbb{R}$) are real then
 (a) $\lambda > \frac{4}{3}$ (b) $\lambda > \frac{5}{3}$ (c) $\lambda \in \left(\frac{1}{3}, \frac{5}{3}\right)$ (d) $\lambda \in \left(\frac{4}{3}, \frac{5}{3}\right)$
- Q 38. If $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & -2 & 4 \end{bmatrix}$ and $A^{-1} = \frac{1}{6}(A^2 + cA + dI)$, where I is the identity matrix, then the value of (c, d) is
 (a) $(-6, -11)$ (b) $(6, 11)$ (c) $(-6, 11)$ (d) $(6, -11)$
- Q 39. Let $f''(x) = -f(x)$ and $g(x) = f'(x)$. If $F(x) = \left\{ f\left(\frac{x}{2}\right)^2 \right\} \left\{ g\left(\frac{x}{2}\right) \right\}^2$ and $F(5) = 5$ then $F(10)$ is equal to
 (a) 5 (b) 10 (c) 0 (d) 15
- Q 40. Let $0 = \left(0, \frac{\pi}{4}\right)$ and $t_1 = (\tan \theta)^{\tan \theta}$, $t_2 = (\tan \theta)^{\cot \theta}$, $t_3 = (\cot \theta)^{\tan \theta}$ and $t_4 = (\cot \theta)^{\cot \theta}$ then
 (a) $t_1 > t_2 > t_3 > t_4$ (b) $t_4 > t_3 > t_1 > t_2$ (c) $t_3 > t_1 > t_2 > t_4$ (d) $t_2 > t_3 > t_1 > t_4$

One-or-More-Options-Correct Questions

Q 41. Let $\vec{A}$ be a vector parallel to the line of intersection of the planes P_1 and P_2 through the origin. The plane P_1 is parallel to the vectors $2\vec{j} + 3\vec{k}$ and $4\vec{j} - 3\vec{k}$ while the plane P_2 is parallel to the vectors $\vec{j} - \vec{k}$ and $3\vec{i} + 3\vec{j}$. The angle between $\vec{A}$ and $2\vec{i} + \vec{j} - 2\vec{k}$ is

- (a) $\frac{\pi}{2}$ (b) $\frac{\pi}{4}$ (c) $\frac{\pi}{6}$ (d) $\frac{3\pi}{4}$

Q 42. The equations of common tangents to the parabolas $y = x^2$ and $y = -(x - 2)^2$ are
 (a) $y = 0$ (b) $y = 4(x - 1)$ (c) $y + 4(x - 1) = 0$ (d) $y + 30x + 50 = 0$

Q 43. Let $f(x) = \begin{cases} e^x & \text{for } 0 \leq x \leq 1 \\ 2 - e^{x-1} & \text{for } 1 < x \leq 2 \\ x - e & \text{for } 2 < x \leq 3 \end{cases}$

and $g(x) = \int_0^x f(t) dt$, $1 < x < 3$. Then $g(x)$ has

- (a) a local maximum at $x = 1 + \log 2$ and a local minimum at $x = e$
 (b) a local maximum at $x = e$ and a local minimum at $x = 1 + \log 2$
 (c) no local maxima
 (d) no local minima

Comprehension-Type Questions

Q 44. There are n urns each containing $n + 1$ balls such that the i th urn contains i white balls and $(n + 1 - i)$ red balls. Let u_i be the event of selecting i th urn and w denote the event of getting a white ball.

(i) If $P(u_i) = \frac{1}{n+1}$ where $i = 1, 2, 3, \dots, n$ then $\lim_{n \rightarrow \infty} P(w)$ is equal to

- (a) $\frac{2}{3}$ (b) $\frac{3}{4}$ (c) $\frac{1}{4}$ (d) 1

(ii) If $P(u_i) = c$, a constant then, $P\left(\frac{u_n}{w}\right)$ is equal to

- (a) $\frac{1}{n+1}$ (b) $\frac{2}{n+1}$ (c) $\frac{n}{n+1}$ (d) $\frac{1}{2}$

(iii) If n is even and E denotes the event of choosing even-numbered urn and u_i s are equiprobable then the value of $P\left(\frac{w}{E}\right)$ is

- (a) $\frac{1}{n+1}$ (b) $\frac{n}{n+1}$ (c) $\frac{n+2}{2(n+1)}$ (d) $\frac{n+2}{2n+1}$

Matching Questions

More than one may match with the same.

Q 45. Normals are drawn at points P, Q and R lying on the parabola $y^2 = 4x$ and they intersect at (3, 0). Then

- (i) Area of ΔPQR (a) 2
 (ii) Circumradius of ΔPQR (b) $\frac{5}{2}$
 (iii) Centroid of ΔPQR (c) $\left(\frac{5}{2}, 0\right)$
 (iv) Circumcentre of ΔPQR (d) $\left(\frac{2}{3}, 0\right)$

Q 46. (i) Two rays in the first quadrant are $x + y = |a|$ and $ax - y = 1$ which intersect each other if $a \in (a_0, \infty)$. The value of a_0 is (a) 2

- (ii) The point (α, β, γ) lies on the plane (b) $\frac{4}{3}$

$$x + y + z = 2.$$

If $\vec{\alpha} = \alpha \vec{i} + \beta \vec{j} + \gamma \vec{k}$ and $\vec{k} \times (\vec{k} \times \vec{a}) = 0$ then γ is

$$(c) \left| \int_0^1 \sqrt{1-x} \, dx \right| + \left| \int_{-1}^0 \sqrt{1+x} \, dx \right|$$

(iv) If in a $\triangle ABC$,
 $\sin A \sin B \sin C + \cos A \cos B = 1$ then
the value of $\sin C$ is

(d) 1

Complete the following statement.

Q 47. The value of $5050 \times \frac{\int_1^0 (1-x^{50})^{100} dx}{\int_0^1 (1-x^{50})^{101} dx}$ is

Answers

1a	2b	3d	4c	5a	6b	7c	8c	9d	10d
11d	12d	13c	14a	15a	16b	7c	18b	19a	20d
21c	22c	23b	24a	25d	26b	27d	28b	29c	30d
31a	32b	33c	34b	35a	36d	37a	38c	39a	40b
41bd	43ab	43a							
44.	(i) a	(ii) b	(iii) c						
45.	(i) a	(ii) b	(iii) bc	(iv) c					
46.	(i) d	(ii) a	(iii) bc	(iv) d					
47.	5051								

IIT Questions

One–Option–Correct Questions

- Q 1. Let $\omega = -\frac{1}{2} + i\frac{\sqrt{3}}{2}$. Then the value of the determinant $\begin{vmatrix} 1 & 1 & 1 \\ 1 & -1-\omega^2 & \omega^2 \\ 1 & \omega^2 & \omega^4 \end{vmatrix}$ is
- (a) 3ω (b) $3\omega(\omega - 1)$ (c) $3\omega^2$ (d) $3\omega(1 - \omega)$
- Q 2. For all complex numbers z_1, z_2 satisfying $|z_1| = 12$ and $|z_2 - 3 - 4i| = 5$, the minimum value of $|z_1 - z_2|$ is
- (a) 0 (b) 2 (c) 7 (d) 17
- Q 3. If $a_1, a_2, \dots, a_n$ are positive real numbers whose product is a fixed number c , the minimum value of $a_1 + a_2 + \dots + a_{n-1} + 2a_n$ is
- (a) $n(2c)^{1/n}$ (b) $(n + 1)c^{1/n}$ (c) $2nc^{1/n}$ (d) $(n + 1)(2c)^{1/n}$
- Q 4. Suppose a, b, c are in AP and a^2, b^2, c^2 are in GP. If $a < b < c$ and $a + b + c = \frac{3}{2}$ then the value of a is
- (a) $\frac{1}{2\sqrt{2}}$ (b) $\frac{1}{2\sqrt{3}}$ (c) $\frac{1}{2} - \frac{1}{\sqrt{3}}$ (d) $\frac{1}{2} - \frac{1}{\sqrt{2}}$
- Q 5. The number of arrangements of the letters of the word BANANA in which two Ns do not appear adjacently is
- (a) 40 (b) 60 (c) 80 (d) 100
- Q 6. The sum $\sum_{i=0}^m \binom{10}{i} \binom{20}{m-i} \cdot \binom{p}{q} = 0$ if $p < q$, is the maximum when m is
- (a) 5 (b) 10 (c) 15 (d) 20
- Q 7. The number of values of k for which the system of equations
- $$\begin{aligned} (k + 1)x + 8y &= 4k, \\ k.x + (k + 3)y &= 3k - 1 \end{aligned}$$
- has infinitely many solutions is
- (a) 0 (b) 1 (c) 2 (d) infinite
- Q 8. The set of all real x for which $x^2 - |x + 2| + x > 0$ is
- (a) $(-\infty, -2) \cup (2, \infty)$ (b) $(-\infty, -\sqrt{2}) \cup (\sqrt{2}, \infty)$ (c) $(-\infty, -1) \cup (1, \infty)$ (d) $\sqrt{2}, \infty)$
- Q 9. The length of the longest interval in which the function $3\sin x - 4\sin^3 x$ is increasing is
- (a) $\frac{\pi}{3}$ (b) $\frac{\pi}{2}$ (c) $\frac{3\pi}{2}$ (d) π
- Q 10. Which of the following pieces of data does not uniquely determine an acute-angled $\triangle ABC$ (R = circumradius)?
- (a) $a, \sin A, \sin B$ (b) a, b, c (c) $a, \sin B, R$ (d) $a, \sin A, R$
- Q 11. The number of integral values of k for which the equation $7\cos x + 5\sin x = 2k + 1$ has a solution is
- (a) 4 (b) 8 (c) 10 (d) 12
- Q 12. Let $0 < \alpha < \frac{\pi}{2}$ be a fixed angle. If $P = (\cos \theta, \sin \theta)$ and $Q = (\cos(\alpha - \theta), \sin(\alpha - \theta))$ then Q is obtained from P by
- (a) clockwise rotation around origin through an angle α
 (b) anticlockwise rotation around origin through an angle α
 (c) reflection in the line through origin with slope $\tan \alpha$
 (d) reflection in the line through origin with slope $\tan \frac{\alpha}{2}$
- Q 13. Let $P = (-1, 0), Q = (0, 0)$ and $R = (3, 3\sqrt{3})$ be three points. Then the equation of the bisector of the angle PQR is

- (a) $\frac{\sqrt{3}}{2}x + y = 0$ (b) $x + \sqrt{3}y = 0$ (c) $\frac{\sqrt{3}}{2}x + y = 0$ (d) $x + \frac{\sqrt{3}}{2}y = 0$
- Q 14. A straight line through the origin O meets the parallel lines $4x + 2y = 9$ and $2x + y + 6 = 0$ at points P and Q, respectively. Then the point O divides the segment PQ in the ratio
(a) 1 : 2 (b) 3 : 4 (c) 2 : 1 (d) 4 : 3
- Q 15. If the tangent at the point P on the circle $x^2 + y^2 + 6x + 6y = 2$ meets the straight line $5x - 2y + 6 = 0$ at a point Q on the y-axis then the length of PQ is
(a) 4 (b) $2\sqrt{5}$ (c) 5 (d) $3\sqrt{5}$
- Q 16. If $a > 2b > 0$ then the positive value of m for which $y = mx - b\sqrt{1+m^2}$ is a common tangent to $x^2 + y^2 = b^2$ and $(x-a)^2 + y^2 = b^2$ is
(a) $\frac{2b}{\sqrt{a^2 - 4b^2}}$ (b) $\frac{\sqrt{a^2 - 4b^2}}{2b}$ (c) $\frac{2b}{a - 2b}$ (d) $\frac{b}{a - 2b}$
- Q 17. The locus of the midpoint of the line segment joining the focus to a moving point on the parabola $y^2 = 4ax$ is another parabola with the directrix
(a) $x = -a$ (b) $x = -\frac{a}{2}$ (c) $x = 0$ (d) $x = \frac{a}{2}$
- Q 18. The area bounded by the curves $y = |x| - 1$ and $y = -|x| + 1$ is
(a) 1 (b) 2 (c) $2\sqrt{2}$ (d) 4
- Q 19. Suppose $f(x) = (x+1)^2$ for $x > -1$. If $g(x)$ is the function whose graph is the reflection of the graph of $f(x)$ with respect to the line $y = x$ then $g(x)$ equals
(a) $-\sqrt{x} - 1$ for $x \geq 0$ (b) $\frac{1}{(x+1)^2}$ for $x > -1$ (c) $\sqrt{x+1}$ for $x \geq -1$ (d) $\sqrt{x} - 1$ for $x \geq 0$
- Q 20. Let the function $f: \mathbb{R} \rightarrow \mathbb{R}$ be defined by $f(x) = 2x + \sin x$ ($x \in \mathbb{R}$). Then f is
(a) one-to-one and onto (b) one-to-one but not onto
(c) onto but not one-to-one (d) neither one-to-one nor onto
- Q 21. The domain of the derivative of the function

$$f(x) = \begin{cases} \tan^{-1} x & \text{for } |x| \leq 1 \\ \frac{1}{2}(|x| - 1) & \text{for } |x| > 1 \end{cases}$$
is
(a) $\mathbb{R} - \{0\}$ (b) $\mathbb{R} - \{1\}$ (c) $\mathbb{R} - \{-1\}$ (d) $\mathbb{R} - \{-1, 1\}$
- Q 22. The integer n for which $\lim_{x \rightarrow 0} \frac{(\cos x - 1)(\cos x - e^x)}{x^n}$ is a finite nonzero number is
(a) 1 (b) 2 (c) 3 (d) 4
- Q 23. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be such that $f(1) = 3$ and $f'(1) = 6$. Then $\lim_{x \rightarrow 0} \left\{ \frac{f(1+x)}{f(1)} \right\}^{1/x}$ equals
(a) 1 (b) $e^{1/2}$ (c) e^2 (d) e^3
- Q 24. The point(s) on the curve $y^3 + 3x^2 = 12y$, where the tangent is vertical (i.e., parallel to the y-axis), is/are
(a) $\left(\pm \frac{4}{\sqrt{3}}, -2 \right)$ (b) $\left(\pm \frac{\sqrt{11}}{3}, 1 \right)$ (c) (0, 0) (d) $\left(\pm \frac{4}{\sqrt{3}}, 2 \right)$
- Q 25. The equation of the common tangent to the curves $y^2 = 8x$ and $xy = -1$ is
(a) $3y = 9x + 2$ (b) $y = 2x + 1$ (c) $2y = x + 8$ (d) $y = x + 2$
- Q 26. Let $f(x) = \int_1^x \sqrt{2-t^2} dt$. Then real roots of the equation $x^2 - f'(x) = 0$ are
(a) ± 1 (b) $\pm \frac{1}{\sqrt{2}}$ (c) $\pm \frac{1}{2}$ (d) 0 and 1
- Q 27. Let $T > 0$ be a fixed number. Suppose f is continuous function such that for all $x \in \mathbb{R}$, $f(x+T) = f(x)$. If $I = \int_0^T f(x) dx$ then the value of $\int_3^{3+3T} f(2x) dx$ is

- (a) $\frac{3}{2}I$ (b) $2I$ (c) $3I$ (d) $6I$
- Q 28. The integral $\int_{-1/2}^{1/2} \left\{ [x] + \log \left(\frac{1+x}{1-x} \right) \right\} dx$ equals
 (a) $-\frac{1}{2}$ (b) 0 (c) 1 (d) $2\log \frac{1}{2}$
- Q 29. If $\vec{a}$ and $\vec{b}$ are two unit vectors such that $\vec{a} + 2\vec{b}$ and $5\vec{a} - 4\vec{b}$ are perpendicular to each other then the angle between $\vec{a}$ and $\vec{b}$ is
 (a) 45° (b) 60° (c) $\cos^{-1} \left(\frac{1}{3} \right)$ (d) $\cos^{-1} \left(\frac{2}{7} \right)$
- Q 30. Let $\vec{V} = 2\vec{i} + \vec{j} - \vec{k}$ and $\vec{W} = \vec{i} + 3\vec{k}$. If $\vec{U}$ is a unit vector then the maximum value of the scalar triple product $[\vec{U} \vec{V} \vec{W}]$ is
 (a) -1 (b) $\sqrt{10} + \sqrt{6}$ (c) $\sqrt{59}$ (d) $\sqrt{60}$
- Q 31. If $\theta \in (0, 2\pi)$ then the interval of values of θ for which $2\sin^2\theta - 5\sin\theta + 2 > 0$ is
 (a) $\left(\frac{\pi}{8}, \frac{5\pi}{6} \right)$ (b) $\left(\frac{41\pi}{48}, \pi \right)$ (c) $\left(0, \frac{\pi}{6} \right) \cup \left(\frac{5\pi}{6}, 2\pi \right)$ (d) $\left(0, \frac{\pi}{8} \right) \cup \left(\frac{\pi}{6}, \frac{5\pi}{6} \right)$
- Q 32. If w is a nonreal complex number such that $\frac{w - \bar{w}z}{1 - z}$ is purely real then the set of values of z is
 (a) $\{z : z = \bar{z}\}$ (b) $\{z : |z| = 1\}$ (c) $\{z : z \in \mathbb{N}\}$ (d) $\{z : |z| = 1, z \neq 1\}$
- Q 33. If r, s and t are prime numbers and p, q are positive integers such that the LCM of p, q is $r^2 t^4 s^2$ then the number of ordered pair (p, q) is
 (a) 254 (b) 252 (c) 225 (d) 224
- Q 34. The axis of a parabola is along the line $y = x$ and the distance of the origin from its vertex is $\sqrt{2}$ and that from its focus is $2\sqrt{2}$. If the vertex and focus both lie in the first quadrant then the equation of the parabola is
 (a) $(x + y)^2 = x - y - 2$ (b) $(x - y)^2 = x + y - 2$ (c) $(x - y)^2 = 4(x + y - 2)$ (d) $(x - y)^2 = 8(x + y - 2)$

One-or-More-Options-Correct Questions

- Q 35. A tangent to the curve $y = f(x)$ at $P(x, y)$ cuts the x -axis and the y -axis at A and B , respectively, such that $BP:AP = 3:1$. If $f(1) = 1$ then
 (a) the equation of the curve is $x \frac{dy}{dx} + 3y = 0$
 (b) the curve passes through $\left(2, \frac{1}{8} \right)$
 (c) the equation of the curve is $x \frac{dy}{dx} - 3y = 0$
 (d) the normal at $(1, 1)$ is $x + 3y = 4$
- Q 36. The internal bisector of the angle A of the $\triangle ABC$ meets BC at D . A line drawn through D perpendicular to AD intersects the side AC at E and the side AB produced at F . Then
 (a) HM of b and c is equal to AE (b) the $\triangle AEF$ is isosceles
 (c) $AD = \frac{2bc}{b+c} \cos \frac{A}{2}$ (d) $EF = \frac{4bc}{b+c} \sin \frac{A}{2}$
- Q 37. $f(x)$ is a polynomial of the third degree which has a local maximum at $x = -1$. If $f(1) = -1$, $f(2) = 18$ and $f'(x)$ has a local minimum at $x = 0$ then
 (a) $f(0) = 5$
 (b) $f(x)$ has a local minimum at $x = 1$
 (c) $f(x)$ is increasing in $[1, 2\sqrt{5}]$
 (d) the distance between $(-1, 2)$ and $(a, f(a))$, where a is a point of local minimum is $2\sqrt{5}$

Comprehension-Type Questions

Q 38. Suppose we define definite integral using the formula

$$\int_a^b f(x) dx = \frac{b-a}{2} \{f(a) + f(b)\}. \text{ For more accurate result, we have}$$

$$\int_a^b f(x) dx = \frac{b-a}{4} \{f(a) + f(b) + 2f(c)\} \text{ when } c = \frac{a+b}{2}. \text{ Also, let}$$

$$F(c) = \frac{c-a}{2} \{f(a) + f(c)\} + \frac{b-c}{2} \{f(b) + f(c)\} \text{ when } c \in (a, b).$$

(i) $\int_0^{\pi/2} \sin x \, dx$ equals

- (a) $\frac{\pi}{8}(1+\sqrt{2})$ (b) $\frac{\pi}{4}(1+\sqrt{2})$ (c) $\frac{\pi}{8\sqrt{2}}$ (d) $\frac{\pi}{4\sqrt{2}}$

(ii) $\int_a^b f(x) \, dx$ is equal to

- (a) $F\left(\frac{a-b}{2}\right)$ (b) $F\left(\frac{a+b}{2}\right)$ (c) $F\left(\frac{a-b}{4}\right)$ (d) $F\left(\frac{a+b}{2}\right)$

(iii) If $f''(x) < 0 \forall x \in (a, b)$ and $(c, f(c))$ is a point lying on the curve $y = f(x)$, where $a < c < b$ and for that value of c , $F(c)$ has a maximum, then $f'(c)$ equals

- (a) $\frac{f(b)-f(a)}{b-a}$ (b) $\frac{2}{b-a}\{f(b)-f(a)\}$ (c) $\frac{2f(b)-f(a)}{2b-a}$ (d) 0

Matching Questions

More than one may match with the same.

Q 39. (i) If $\sum_{n=1}^{\infty} \tan^{-1} \frac{1}{2n^2} = t$ then $\tan t$ equals (a) 0

(ii) If in the $\triangle ABC$ the sides a, b, c are in (b) 1

$$AP \text{ and } \cos \theta_1 = \frac{a}{b+c}, \cos \theta_2 = \frac{b}{a+c}$$

$$\text{and } \cos \theta_3 = \frac{c}{a+b} \text{ then } \tan^2 \frac{\theta_1}{2} \tan^2 \frac{\theta_3}{2} \text{ equals}$$

(iii) A line perpendicular to $x + 2y + 2z = 0$ and (c) $\frac{\sqrt{5}}{3}$

passes through $(0, 1, 0)$. The perpendicular distances of the from the origin is

(iv) $\int_0^1 e^{x^2} dx + \int_1^e (-1) + \sqrt{\log x} \, dx$ has the value (d) $\frac{2}{3}$

Complete the following statement.

Q 4. If $f(x)$ is a twice differentiable function such that $f(a) = 0$, $f(b) = 2$, $f(c) = -1$, $f(d) = 2$ and $f(e) = 0$, where $a < b < c < d < e$, then the minimum number of zero of $g(x) = \{f'(x)\}^2 + f''(x) \cdot f(x)$ in the interval $[a, e]$ is

Answers

1b	2b	3a	4d	5a	6a	7b	8b	9a	10d
11b	12d	13c	14b	15c	16a	17c	18b	19d	20a
21c	22c	23c	24d	25d	26a	27c	28a	29b	30c
31c	32d	33d	34d	35ab					

36. a,b,c,d

37. b,c

38. (i) a (ii) b (iii) a

39. (i) b (ii) d (iii) c (iv) b

40. 6

IIT Questions

Straight Objective Type (One–Option–Correct Questions)

- Q 1. The number of solutions of the pair of equation

$$2\sin^2 \theta - \cos 2\theta = 0$$

$$2\cos^2 \theta - 3\sin \theta = 0$$
in the interval $[0, 2\pi]$ is
(a) zero (b) one (c) two (d) four
- Q 2. Let α, β be the roots of the equation $x^2 - px + r = 0$ and $\frac{\alpha}{2}, 2\beta$ be the roots of the equation $xr - qx + r = 0$. Then the value of r is
(a) $\frac{2}{9}(p - q)(2q - p)$ (b) $\frac{2}{9}(q - p)(2p - q)$ (c) $\frac{2}{9}(q - 2p)(2q - p)$ (d) $\frac{2}{9}(2p - q)(2q - p)$
- Q 3. The number of distinct real values of X , for which the vectors $-\lambda^2 i + j + k, i - \lambda^2 j + k$ and $i + j - \lambda^2 k$ are coplanar, is
(a) zero (b) one (c) two (d) three
- Q 4. One Indian and four American men and their wives are to be seated randomly around a circular table. Then the conditional probability that the Indian man is seated adjacent to his wife given that each American man is seated adjacent to his wife is
(a) $\frac{1}{2}$ (b) $\frac{1}{3}$ (c) $\frac{2}{5}$ (d) $\frac{1}{5}$
- Q 5. $\lim_{x \rightarrow 0} \frac{\int_0^{\sec^2 x} f(t) dt}{x^2 - \frac{\pi^2}{16}}$ equals
(a) $\frac{8}{\pi} f(2)$ (b) $\frac{2}{\pi} f(2)$ (c) $\frac{2}{\pi} f\left(\frac{1}{2}\right)$ (d) $4f(2)$
- Q 6. Let $f(x) = \frac{x}{(1+x^n)^{1/n}}$ for $n \geq 2$ and $g(x) = \underbrace{(f \circ f \circ \dots \circ f)}_{f \text{ occurs } n \text{ times}}(x)$. Then $\int (x) dx$ equals
(a) $\frac{1}{n(n-1)}(1+nx^n)^{\frac{1}{n}-1} + K$ (b) $\frac{1}{n-1}(1+nx^n)^{\frac{1}{n}-1} + K$
(c) $\frac{1}{n(n+1)}(1+nx^n)^{\frac{1}{n}+1} + K$ (d) $\frac{1}{n+1}(1+nx^n)^{\frac{1}{n}+1} + K$
- Q 7. The letters of the word COCHIN are permuted and all the permutations are arranged in an alphabetical order as in an English dictionary. The number of words that appear before the word COCHIN is
(a) 360 (b) 192 (c) 96 (d) 48
- Q 8. Let ABCD be a quadrilateral with area 18, with side AB parallel to the side CD and $AB = 2CD$. Let AD be perpendicular to AB and CD. If a circle is drawn inside the quadrilateral ABCD touching all the sides, then its radius is
(a) 3 (b) 2 (c) $\frac{3}{2}$ (d) 1
- Q 9. A hyperbola, having the transverse axis of length $2\sin \theta$, is confocal with the ellipse $3x^2 + 4y^2 = 12$. Then its equation is
(a) $x^2 \operatorname{cosec}^2 \theta - y^2 \sec^2 \theta = 1$ (b) $x^2 \sec^2 \theta - y^2 \operatorname{cosec}^2 \theta = 1$
(c) $x^2 \sin^2 \theta - y^2 \cos^2 \theta = 1$ (d) $x^2 \cos^2 \theta - y^2 \sin^2 \theta = 1$
- Q 10. The tangent to the curve $y = e^x$ drawn at the point (c, e^c) intersects the line joining the points $(c - 1, e^{c-1})$ and $(c + 1, e^{c+1})$
(a) on the left of $x = c$ (b) on the right of $x = c$ (c) at no point (d) at all points

- Q 11. A man walks a distance of 3 units from the origin towards the north-east ($N 45^\circ E$) direction. From there, he walks a distance of 4 units towards the north-west ($N 45^\circ W$) direction to reach a point P. Then the position of P in the Argand plane is
 (a) $3e^{i\pi/4} + 4i$ (b) $(3 - 4i)^{i\pi/4}$ (c) $(4 + 3i)e^{i\pi/4}$ (d) $(3 + 4i)e^{i\pi/4}$
- Q 12. Let $f(x)$ be differentiable on the interval $(0, \infty)$ such that $f(1) = 1$, and $\lim_{t \rightarrow x} \frac{t^2 f(x) - x^2 f(t)}{t - x} = 1$ for each $x > 0$. The $f(x)$ is
 (a) $\frac{1}{3x} + \frac{2x^2}{3}$ (b) $\frac{-1}{3x} + \frac{4x^2}{3}$ (c) $\frac{-1}{x} + \frac{2}{x^2}$ (d) $\frac{1}{x}$
- Q 13. $\frac{d^2x}{dy^2}$ equals
 (a) $\left(\frac{d^2y}{dx^2}\right)^{-1}$ (b) $\left(\frac{d^2y}{dx^2}\right)^{-1} \left(\frac{dx}{dy}\right)^{-3}$ (c) $\left(\frac{d^2y}{dx^2}\right) \left(\frac{dy}{dx}\right)^{-2}$ (d) $-\left(\frac{d^2y}{dx^2}\right) \left(\frac{dy}{dx}\right)^{-3}$
- Q 14. The differential equation $\frac{dy}{dx} \frac{\sqrt{1-y^2}}{y}$ determines a family of circles with
 (a) variable radii and a fixed centre at $(0, 1)$
 (b) variable radii and a fixed centre at $(0, -1)$
 (c) fixed radius 1 and variable centres along the x-axis
 (d) fixed radius 1 and variable centres along the y-axis
- Q 15. If $|z| = 1$ and $z \neq \pm 1$ then all the values of $\frac{z}{1-z^2}$ lie on
 (a) a line not passing through the origin
 (b) $|z| = \sqrt{2}$
 (c) the x-axis
 (d) the y-axis
- Q 16. Let $O(0, 0)$, $P(3, 4)$, $Q(6, 0)$ be the vertices of the triangle OPQ. The point R inside the triangle OPQ is such that the triangles OPR, PQR, OQR are of equal area. The coordinates of R are
 (a) $\left(\frac{4}{3}, 3\right)$ (b) $\left(3, \frac{2}{3}\right)$ (c) $\left(3, \frac{4}{3}\right)$ (d) $\left(\frac{4}{3}, \frac{2}{3}\right)$

Assertion-Reason Type (One-Option-Correct Questions)

- Q 17. Let $F(x)$ be an indefinite integral of $\sin^2 x$.
 STATEMENT-1: The function $F(x)$ satisfies $F(x + n) = F(x)$ for all real x . because
 STATEMENT-2: $\sin^2(x + n) = \sin^2 x$ for all real x .
 (a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1
 (b) Statement-1 is True, Statement-2 is True; Statement-2 is Not a correct explanation for Statement-1
 (c) Statement-1 is True, Statement-2 is False
 (d) Statement-1 is False, Statement-2 is True
- Q 18. Consider the planes $3x - 6y - 2z = 15$ and $2x + y - 2z = 5$.
 STATEMENT-1: The parametric equations of the line of intersection of the given planes are $x = 3 + 14t$, $y = 1 + 2t$, $z = 15t$. because
 STATEMENT-2: The vector $14i + 2j + 15k$ is parallel to the line of intersection of given planes.
 (a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1
 (b) Statement-1 is True, Statement-2 is True; Statement-2 is Not a correct explanation for Statement-1
 (c) Statement-1 is True, Statement-2 is False
 (d) Statement-1 is False, Statement-2 is True

- Q 19. STATEMENT-1: The curve $y = \frac{-x^2}{2} + x + 1$ is symmetric with respect to the line $x = 1$.
because
 STATEMENT-2: A parabola is symmetric about its axis.
 (a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1
 (b) Statement-1 is True, Statement-2 is True; Statement-2 is Not a correct explanation for Statement-1
 (c) Statement-1 is True, Statement-2 is False
 (d) Statement-1 is False, Statement-2 is True
- Q 20. Let $H_1, H_2, H_3, \dots, H_n$ be mutually exclusive and exhaustive events with $P(H_i) > 0$; $i = 1, 2, 3, \dots, n$. Let E be any other event with $0 < P(E) < 1$.
 STATEMENT-1: $P(H_i | E) > P(E | H_i) \cdot P(H_i)$ for $i = 1, 2, 3, \dots, n$.
because
 STATEMENT-2: $\sum_{i=1}^n P(H_i) = 1$.
 (a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1
 (b) Statement-1 is True, Statement-2 is True; Statement-2 is Not a correct explanation for Statement-1
 (c) Statement-1 is True, Statement-2 is False
 (d) Statement-1 is False, Statement-2 is True
- Q 21. Tangents are drawn from the point $(17, 7)$ to the circle $x^2 + y^2 = 169$.
 STATEMENT-1: The tangents are mutually perpendicular.
because
 STATEMENT-2: The locus of the points from which mutually perpendicular tangents can be drawn to the given circle is $x^2 + y^2 = 338$.
 (a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1
 (b) Statement-1 is True, Statement-2 is True; Statement-2 is Not a correct explanation for Statement-1
 (c) Statement-1 is True, Statement-2 is False
 (d) Statement-1 is False, Statement-2 is True
- Q 22. Let $f(x) = 2 + \cos x$ for all real x .
 STATEMENT-1: For each real t , there exists a point c in $[t, t + \pi]$ such that $f'(c) = 0$.
because
 STATEMENT-2: $f(t) = f(t + 2\pi)$ for each real t .
 (a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1
 (b) Statement-1 is True, Statement-2 is True; Statement-2 is Not a correct explanation for Statement-1
 (c) Statement-1 is True, Statement-2 is False
 (d) Statement-1 is False, Statement-2 is True
- Q 23. Lines $L_1 : y - x = 0$ and $L_2 : 2x + y = 0$ intersect the line $L_3 : y + 2 = 0$ at P and Q respectively. The bisector of the acute angle between L_1 and L_2 intersects L_3 at R .
 STATEMENT-1: The ratio $PR : RQ = 2 : 2 : \sqrt{5}$.
because
 STATEMENT-2: In any triangle, bisector of an angle divides the triangle into two similar triangles.
 (a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1
 (b) Statement-1 is True, Statement-2 is True; Statement-2 is Not a correct explanation for Statement-1
 (c) Statement-1 is True, Statement-2 is False
 (d) Statement-1 is False, Statement-2 is True

Linked Comprehension Type (One-Option-Correct Questions) Paragraph for Question Nos. 24 to 26

Let A_1, G_1, H_1 denote the arithmetic, geometric and harmonic means, respectively, of two distinct positive numbers. For $n > 2$, let A_{n-1} and H_{n-1} have arithmetic, geometric and harmonic means as A_n, G_n, H_n respectively.

Q 24. Which one of the following statements is correct?

- (a) $G_1 > G_2 > G_3 > \dots$
- (b) $G_1 < G_2 < G_3 < \dots$
- (c) $G_1 = G_2 = G_3 = \dots$
- (d) $G_1 < G_3 < G_5 < \dots$ and $G_2 > G_4 > G_6 > \dots$

Q 25. Which one of the following statements is correct?

- (a) $A_1 > A_2 > A_3 > \dots$
- (b) $A_1 < A_2 < A_3 < \dots$
- (c) $A_1 > A_3 > A_5 > \dots$ and $A_2 < A_4 < A_6 < \dots$
- (d) $A_1 < A_3 < A_5 < \dots$ and $A_2 > A_4 > A_6 > \dots$

Q 26. Which one of the following statements is correct?

- (a) $H_1 > H_2 > H_3 > \dots$
- (b) $H_1 < H_2 < H_3 < \dots$
- (c) $H_1 > H_3 > H_5 > \dots$ and $H_2 < H_4 < H_6 < \dots$
- (d) $H_1 < H_3 < H_5 < \dots$ and $H_2 > H_4 > H_6 > \dots$

Paragraph for Question Nos. 27 to 29

Consider the circle $x^2 + y^2 = 9$ and the parabola $y^2 = 8x$. They intersect at P and Q in the first and the fourth quadrants, respectively. Tangents to the circle at P and Q intersect the x-axis at R and tangents to the parabola at P and Q intersect the x-axis at S.

Q 27. The ratio of the areas of the triangles PQS and PQR is

- (a) $1 : \sqrt{2}$
- (b) $1 : 2$
- (c) $1 : 4$
- (d) $1 : 8$

Q 28. The radius of the circumcircle of the triangle PRS is

- (a) 5
- (b) $3\sqrt{3}$
- (c) $3\sqrt{2}$
- (d) $2\sqrt{3}$

Q 29. The radius of the incircle of the triangle PQR is

- (a) 4
- (b) 3
- (c) $\frac{8}{3}$
- (d) 2

Paragraph for Question Nos. 30 to 32

Let V_r denote the sum of the first r terms of an arithmetic progression (AP) whose first term is r and the common difference is $(2r-1)$. Let

$T_r = V_{r+1} - V_r - 2$ and $Q_r = T_{r+1} - T_r$ for $r = 1, 2, \dots$

Q 30. The sum $V_1 + V_2 + \dots + V_n$ is

- (a) $\frac{1}{12} n(n+1)(3n^2 - n + 1)$
- (b) $\frac{1}{12} n(n+1)(3n^2 + n + 2)$
- (c) $\frac{1}{2} n(2n^2 - n + 1)$
- (d) $\frac{1}{3} (2n^3 - 2n + 3)$

Q 31. T_r is always

- (a) an odd number
- (b) an even number
- (c) a prime number
- (d) a composite number

Q 32. Which one of the following is the correct statement?

- (a) $Q_1, Q_2, Q_3, \dots$ are in AP with common difference 5
- (b) $Q_1, Q_2, Q_3, \dots$ are in AP with common difference 6
- (c) $Q_1, Q_2, Q_3, \dots$ are in AP with common difference 11
- (d) $Q_1 = Q_2 = Q_3 = \dots$

Matrix-Match Type

Statements (A, B, C, D) in Column I have to be matched with statements (p, q, r, s) in Column II. The answers to these questions have to be appropriately bubbled as illustrated in the following example.

	p	q	r	s
A	☒	☐	☐	☒
B	☐	☒	☒	☐
C	☒	☒	☐	☐
D	☐	☐	☐	☒

If the correct matches are A-p, A-s, B-q, B-r, C-p, C-q and D-s then the correctly bubbled 4 x 4 matrix should be as given above.

- Q 33. Match the integrals in Column I with the values in Column II and indicate your answer by darkening the appropriate bubbles in the 4 x 4 matrix

Column I

(A) $\int_{-1}^1 \frac{dx}{1+x^2}$

(B) $\int_{-1}^1 \frac{dx}{\sqrt{1-x^2}}$

(C) $\int_2^3 \frac{dx}{1-x^2}$

(D) $\int_1^2 \frac{dx}{x\sqrt{x^2-1}}$

Column II

(p) $\frac{1}{2} \log\left(\frac{2}{3}\right)$

(q) $2 \log\left(\frac{2}{3}\right)$

(r) $\frac{\pi}{3}$

(s) $\frac{\pi}{2}$

- Q 34. In the following [x] denotes the greatest integer less than or equal to x. Match the functions in Column I with the properties in Column II and indicate your answer by darkening the appropriate bubbles in the 4 x 4 matrix.

Column I

(A) $x|x|$

(B) $\sqrt{|x|}$

(C) $x + [x]$

(D) $|x-1| + |x+1|$

Column II

(p) continuous in $(-1,1)$

(q) differentiable in $(-1,1)$

(r) strictly increasing in $(-1,1)$

(s) not differentiable at least at one point in $(-1,1)$

- Q 35. Match the statement in Column I with the properties in Column II and indicate your answer by darkening the appropriate bubbles in the 4 x 4 matrix.

Column I

(A) Two intersecting circles

(B) Two mutually external circles

(C) Two circles, one strictly inside the other

(D) Two branches of a hyperbola

Column II

(p) have a common tangent

(q) have a common normal

(r) do not have a common tangent

(s) do not have a common normal

- Q 36. Consider the following linear equations

$$ax + by + cz = 0$$

$$bx + cy + az = 0$$

$$cx + ay + bz = 0$$

Match the conditions/expressions in Column I with statements in Column II and indicate your answer by darkening the appropriate bubbles in the 4 x 4 matrix given in the ORS.

Column I

(A) $a + b + c \neq 0$ and $a^2 + b^2 + c^2 = ab + bc + ca$

(B) $a + b + c = 0$ and $a^2 + b^2 + c^2 \neq ab + bc + ca$

(C) $a + b + c \neq 0$ and $a^2 + b^2 + c^2 \neq ab + bc + ca$

(D) $a + b + c = 0$ and $a^2 + b^2 + c^2 = ab + bc + ca$

Column II

(p) the equations represent planes meeting only at a single point.

(q) the equations represent the line $x = y = z$.

(r) the equations represent identical planes.

(s) the equations represent the whole of the three dimensional space.

Q 37. Let (x, y) be such that

$$\sin^{-1}(ax) + \cos^{-1}(y) + \cos^{-1}(bxy) = \frac{\pi}{2}.$$

Match the statements in Column I with statements in Column II and indicate your answer by darkening appropriate bubbles in the 4 x 4 matrix.

Column I

Column II

(A) If $a = 1$ and $b = 0$ then (x, y) (p) lies on the circle $x^2 + y^2 = 1$

(B) If $a = 1$ and $b = 1$ then (x, y) (q) lies on $(x^2 - 1)(y^2 - 1) = 0$

(C) If $a = 1$ and $b = 2$ then (x, y) (r) lies on $y = x$

(D) If $a = 2$ and $b = 2$ then (x, y) (s) lies on $(4x^2 - 1)(y^2 - 1) = 0$

Q 38. Let $f(x) = \frac{x^2 - 6x + 5}{x^2 - 5x + 6}$

Match the expression/statements in Column I with expression/ statements in Column II and indicate your answer by darkening the appropriate bubbles in the 4 x 4 matrix.

Column I

Column II

(A) If $-1 < x < 1$ then $f(x)$ satisfies

(p) $0 < f(x) < 1$

(B) If $1 < x < 2$ then $f(x)$ satisfies

(q) $f(x) < 0$

(C) If $3 < x < 5$ then $f(x)$ satisfies

(r) $f(x) > 0$

(D) If $x > 5$ then $f(x)$ satisfies

(s) $f(x) < 1$

Answers

1c	2d	3c	4c	5a	6a	7c	8b	9a	10a
11d	12a	13d	14c	15d	16c	17d	18d	19a	20b
21a	22a	23c	24c	25a	26b	27c	28b	29d	30b

33.

	p	q	r	s
A	☐	☐	☐	☒
B	☐	☐	☐	☒
C	☒	☐	☐	☐
D	☐	☐	☒	☐

(i.e., $A \rightarrow s, B \rightarrow s,$
 $C \rightarrow p, D \rightarrow r$)

34.

	p	q	r	s
A	☒	☒	☒	☐
B	☒	☐	☐	☒
C	☐	☐	☒	☒
D	☒	☒	☐	☐

(i.e., $A \rightarrow p, A \rightarrow q, A \rightarrow r,$
 $B \rightarrow p, B \rightarrow s, C \rightarrow r,$
 $C \rightarrow s, D \rightarrow p, D \rightarrow q$)

35.

	p	q	r	s
A	☒	☒	☐	☐
B	☒	☒	☐	☐
C	☐	☒	☒	☐
D	☐	☒	☒	☐

(i.e., $A \rightarrow p, A \rightarrow q,$
 $B \rightarrow p, B \rightarrow q,$
 $C \rightarrow q, C \rightarrow r,$
 $D \rightarrow q, D \rightarrow r$)

36.

	p	q	r	s
A	☐	☐	☒	☐
B	☐	☒	☐	☐
C	☒	☐	☐	☐
D	☐	☐	☐	☒

(i.e., $A \rightarrow r, B \rightarrow q,$
 $C \rightarrow p, D \rightarrow s$)

37.

	p	q	r	s
A	☒	☐	☐	☐
B	☐	☒	☐	☐
C	☒	☐	☐	☐
D	☐	☐	☐	☒

(i.e., $A \rightarrow p, B \rightarrow q,$
 $C \rightarrow p, D \rightarrow s$)

38.

	p	q	r	s
A	☒	☐	☐	☐
B	☐	☒	☐	☐
C	☐	☒	☐	☐
D	☒	☐	☐	☐

(i.e., $A \rightarrow p, B \rightarrow q,$
 $C \rightarrow q, D \rightarrow p$)

IIT Questions

Straight Objective Type (One-Option-Correct Questions)

- Q 1. The total number of local maxima and local minima of the function is
- $$f(x) = \begin{cases} (2+x)^3, & -3 < x \leq -1 \\ x^{2/3}, & -1 < x < 2 \end{cases}$$
- (a) 0 (b) 1 (c) 2 (d) 3
- Q 2. Let $g(x) = \frac{(x-1)^n}{\log \cos^m(x-1)}$; $0 < x < 2$, m and n are integers, $m \neq 0$, $n > 0$, and let p be the left hand derivative of $|x-1|$ at $x=1$. If $\lim_{x \rightarrow 1^+} g(x) = p$ then
- (a) $n=1, m=r$ (b) $n=1, m=-1$ (c) $n=2, m=2$ (d) $n>2, m=n$
- Q 3. Let $I = \int \frac{e^x}{e^{4x} + e^{2x} + 1} dx$, $J = \int \frac{e^{-x}}{e^{-4x} + e^{-2x} + 1} dx$. Then, for an arbitrary constant c , the value of $J - I$ equals
- (a) $\frac{1}{2} \log \left(\frac{e^{4x} - e^{2x} + 1}{e^{2x} + e^{2x} + 1} \right) + c$ (b) $\frac{1}{2} \log \left(\frac{e^{2x} - e^x + 1}{e^{2x} + e^x + 1} \right) + c$
- (c) $\frac{1}{2} \log \left(\frac{e^{2x} - e^x + 1}{e^{2x} + e^x + 1} \right) + c$ (d) $\frac{1}{2} \log \left(\frac{e^{4x} - e^{2x} + 1}{e^{4x} + e^{2x} + 1} \right) + c$
- Q 4. Let $g(x) = \log f(x)$, where $f(x)$ is twice differentiable positive function on $(0, \infty)$ such that $f(x+1) = xf(x)$. Then for $N = 1, 2, 3, \dots$
- $$g''\left(N + \frac{1}{2}\right) - g''\left(\frac{1}{2}\right) =$$
- (a) $-4 \left\{ 1 + \frac{1}{9} + \frac{1}{25} + \dots + \frac{1}{(2N-1)^2} \right\}$ (b) $\left\{ 1 + \frac{1}{9} + \frac{1}{25} + \dots + \frac{1}{(2N-1)^2} \right\}$
- (c) $-4 \left\{ 1 + \frac{1}{9} + \frac{1}{25} + \dots + \frac{1}{(2N+1)^2} \right\}$ (d) $4 \left\{ 1 + \frac{1}{9} + \frac{1}{25} + \dots + \frac{1}{(2N+1)^2} \right\}$
- Q 5. Let the function $g: (-\infty, \infty) \rightarrow \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ be given by $g(u) = 2 \tan^{-1}(e^u) - \frac{\pi}{2}$. Then g is
- (a) even and is strictly increasing in $(0, \infty)$
 (b) odd and is strictly decreasing in $(-\infty, \infty)$
 (c) odd and is strictly increasing in $(-\infty, \infty)$
 (d) neither even nor odd but is strictly increasing in $(-\infty, \infty)$
- Q 6. The area of the region between the curves $y = \sqrt{\sec x + \tan x}$ and $y = \sqrt{\sec x - \tan x}$ bounded by the lines $x=0$ and $x = \frac{\pi}{4}$ is
- (a) $\int_0^{\sqrt{2}-1} \frac{t}{(1+t^2)\sqrt{1-t^2}} dt$ (b) $\int_0^{\sqrt{2}-1} \frac{4t}{(1+t^2)\sqrt{1-t^2}} dt$
- (c) $\int_0^{\sqrt{2}+1} \frac{4t}{(1+t^2)\sqrt{1-t^2}} dt$ (d) $\int_0^{\sqrt{2}+1} \frac{t}{(1+t^2)\sqrt{1-t^2}} dt$
- Q 7. Consider a branch of the hyperbola $x^2 - 2y^2 - 2x - 4\sqrt{2}y - 6 = 0$ with vertex at the point A. Let B be one end of its latus rectum. If S is the focus of the hyperbola nearest to the point A then $\ar(\triangle ASB)$ is
- (a) $1/\sqrt{2/3}$ (b) $\sqrt{3/2} - 1$ (c) $1 + \sqrt{2/3}$ (d) $\sqrt{3/2} + 1$

- Q 8. Consider the two curves $C_1 : y^2 = 4x$
 $C_2 : x^2 + y^2 - 6x + 1 = 0$.
 Then
 (a) C_1 and C_2 touch each other only at one point
 (b) C_1 and C_2 touch each other exactly at two points
 (c) C_1 and C_2 intersect (but do not touch) at exactly at two points
 (d) C_1 and C_2 neither intersect nor touch each other
- Q 9. The edges of a parallelopiped are of unit length and are parallel to noncoplanar unit vectors $\hat{a}, \hat{b}, \hat{c}$ such that $\hat{a} \cdot \hat{b} = \hat{b} \cdot \hat{c} = \hat{c} \cdot \hat{a} = \frac{1}{2}$.
 Then the volume of the parallelopiped is
 (a) $\frac{1}{\sqrt{2}}$ (b) $\frac{1}{2\sqrt{2}}$ (c) $\frac{\sqrt{3}}{2}$ (d) $\frac{1}{\sqrt{3}}$
- Q 10. An experiment has 10 equally likely outcomes. Let A and B be two non-empty events of the experiment. If A consists of 4 outcomes, the number of outcomes that B must have so that A and B are independent, is
 (a) 2, 4 or 8 (b) 3, 6 or 9 (c) 4 or 8 (d) 5 or 10
- Q 11. If $0 < x < 1$, then $\sqrt{1+x^2} [\{x \cos(\cot^{-1} x) + \sin(\cot^{-1} x)\}^2 - 1]^{1/2}$
 (a) $\frac{x}{\sqrt{1+x^2}}$ (b) x (c) $x\sqrt{1+x^2}$ (d) $\sqrt{1+x^2}$
- Q 12. Let two non-collinear unit vectors $\hat{a}$ and $\hat{b}$ form an acute angle. A point P moves so that at any time t the position vector $\vec{OP}$ (where O is the origin) is given by $\hat{a} \cos t + \hat{b} \sin t$. When P is the farthest from origin O, let M be the length of $\vec{OP}$ and $\hat{u}$ be the unit vector along $\vec{OP}$. Then
 (a) $\hat{u} = \frac{\hat{a} + \hat{b}}{|\hat{a} + \hat{b}|}$ and $M = (1 + \hat{a} \cdot \hat{b})^{1/2}$ (b) $\hat{u} = \frac{\hat{a} - \hat{b}}{|\hat{a} - \hat{b}|}$ and $M = (1 + \hat{a} \cdot \hat{b})^{1/2}$
 (c) $\hat{u} = \frac{\hat{a} + \hat{b}}{|\hat{a} + \hat{b}|}$ and $M = (1 + 2\hat{a} \cdot \hat{b})^{1/2}$ (d) $\hat{u} = \frac{\hat{a} - \hat{b}}{|\hat{a} - \hat{b}|}$ and $M = (1 + 2\hat{a} \cdot \hat{b})^{1/2}$
- Q 13. A particle P starts from the point $z_0 = 1 + 2i$, where $i = \sqrt{-1}$. It moves first horizontally away from origin by 5 units and then vertically away from origin by 3 units to reach a point z_1 . From z_1 the particle moves $\sqrt{2}$ units in the direction of the vector $i+j$ and then moves through an angle $\pi/2$ in anticlockwise direction on a circle with centre at origin, to reach a point z_2 . The point z_2 is given by
 (a) $6 + 7i$ (b) $-7 + 6i$ (c) $7 + 6i$ (d) $-6 + 7i$
- Q 14. Consider three points $P = [-\sin(\beta - \alpha), -\cos \beta]$, $Q = [\cos(\beta - \alpha), \sin \beta]$ and $R = [\cos(\beta - \alpha + \theta), \sin(\beta - \theta)]$, where $\theta < \alpha$, β , $\theta < \frac{\pi}{4}$. Then
 (a) P lies on the line segment RQ (b) Q lies on the line segment PR
 (c) R lies on the line segment QP (d) P, Q, R are non-collinear
- Q 15. Let a, b be non-zero real numbers. The equation $(ax^2 + by^2 + c)(x^2 - 5xy + 6y^2) = 0$ represents
 (a) four straight lines, when $c = 0$ and a, b are of the same sign
 (b) two straight lines and a circle, when $a = b$ and c is of sign opposite to that of a
 (c) two straight lines and a hyperbola, when a and b are of the same sign and c is of sign opposite to that of a
 (d) a circle and an ellipse, when a and b are of the same sign

Multiple Correct Answers Type (One or More-Options-Correct Questions)

- Q 16. Let $f(x)$ be a non-constant twice differentiable function defined on $(-\infty, \infty)$ such that $f(x) = f(1-x)$ and $f'\left(\frac{1}{4}\right) = 0$. Then,
- (a) $f''(x)$ vanishes at least twice on $[0, 1]$
- (b) $f'\left(\frac{1}{2}\right) = 0$
- (c) $\int_{1/2}^{1/2} f\left(x + \frac{1}{2}\right) \cdot \sin x \, dx = 0$
- (d) $\int_0^{1/2} f(t) \cdot e^{\sin \pi t} \, dt = \int_{1/\sqrt{2}}^1 f(1-t) \cdot e^{\sin \pi t} \, dt$
- Q 17. Let $S_n = \sum_{k=1}^n \frac{n}{n^2 + kn + k^2}$ and $\sum_{k=0}^{n-1} \frac{n}{n^2 + kn + k^2}$ for $n = 1, 2, 3, \dots$. Then,
- (a) $S_n < \frac{\pi}{3\sqrt{3}}$ (b) $S_n > \frac{\pi}{3\sqrt{3}}$ (c) $T_n < \frac{\pi}{3\sqrt{3}}$ (d) $T_n > \frac{\pi}{3\sqrt{3}}$
- Q 18. A straight line through the vertex P of the ΔPQR intersects the side QR at S and the circumcircle of the ΔPQR at T. If S is not the centre of the circumcircle, then
- (a) $\frac{1}{PS} + \frac{1}{ST} < \frac{2}{\sqrt{QS \cdot SR}}$ (b) $\frac{1}{PS} + \frac{1}{ST} > \frac{2}{\sqrt{QS \cdot SR}}$
- (c) $\frac{1}{PS} + \frac{1}{ST} < \frac{4}{QR}$ (d) $\frac{1}{PS} + \frac{1}{ST} > \frac{4}{QR}$
- Q 19. Let $P(x_1, y_1)$ and $Q(x_2, y_2)$; where $y_1 < 0, y_2 < 0$ be the end points of the latus rectum of the ellipse $x^2 + 4y^2 = 4$. The equation of parabola with latus rectum PQ are
- (a) $x^2 + 2\sqrt{3}y = 3 + \sqrt{3}$ (b) $x^2 - 2\sqrt{3}y = 3 + \sqrt{3}$
- (c) $x^2 + 2\sqrt{3}y = 3 - \sqrt{3}$ (d) $x^2 - 2\sqrt{3}y = 3 - \sqrt{3}$

Linked Comprehension Type

Paragraph for Question Nos. 20 to 22

Let A, B, C be three sets of complex numbers as given below.

$$A = \{z : \operatorname{Im}(z) \geq 1\}, B = \{z : |z - 2 - i| = 3\}, C = \{z : \operatorname{Re}((1-i)z) = \sqrt{2}\}$$

- Q 20. The number of elements in the set $A \cap B \cap C$ is
- (a) 0 (b) 1 (c) 2 (d) ∞
- Q 21. Let z be any point in $A \cap B \cap C$. Then, $|z + 1 - i|^2 + |z - 5 - i|^2$ lies between
- (a) 25 and 29 (b) 30 and 34 (c) 35 and 39 (d) 40 and 44
- Q 22. Let z be any point in $A \cap B \cap C$ and let w be any point satisfying $|w - 2 - i| < 3$. Then, $|z| - |w| + 3$ lies between
- (a) -6 and 3 (b) -3 and 6 (c) -6 and 6 (d) -3 and 9

Paragraph for Question Nos. 23 to 25

Consider the function $f: \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = \frac{x^2 - ax + 1}{x^2 + ax + 1}$, $a \in (0, 2)$

- Q 23. Which of the following is true?
- (a) $(2+a)^2 \cdot f'(1) + (2-a)^2 \cdot f'(-1) = 0$
- (b) $(2-a)^2 f'(1) - (2+a)^2 f'(-1) = 0$
- (c) $f(1)f(-1) = (2-a)^2$
- (d) $f(1)f(-1) = -(2+a)^2$
- Q 24. Which of the following is true?
- (a) $f(x)$ is decreasing in $(-1, 1)$ and has a local minimum at $x = 1$
- (b) $f(x)$ is increasing in $(-1, 1)$ and has a local maximum at $x = 1$
- (c) $f(x)$ is increasing in $(-1, 1)$ but has neither a local maximum nor a local minimum at $x = 1$
- (d) $f(x)$ is decreasing in $(-1, 1)$ but has neither a local maximum nor a local minimum at $x = 1$
- Q 25. Let $g(x) = f'(x)$, which of the following is true?
- (a) $g'(x)$ is positive on $(-\infty, 0)$ and negative on $(0, \infty)$

- (b) $g'(x)$ is negative on $(-\infty, 0)$ and positive on $(0, \infty)$
 (c) $g'(x)$ changes sign on both $(-\infty, 0)$ and $(0, \infty)$
 (d) $g'(x)$ does not change sign on $(-\infty, \infty)$

Paragraph for Question Nos. 26 to 28

A circle C of radius 1 is inscribed in an equilateral triangle PQR. The points of contact of C with the sides PQ, QR, RP are D, E, F respectively. The line PQ is given by the equation $\sqrt{3}x + y - 6 = 0$ and the point D is $\left(\frac{3\sqrt{3}}{2}, \frac{3}{2}\right)$. Further, it is given that the origin and the centre of C are on the same side of the line PQ.

Q 26. The equation of the circle C is

- (a) $(x - 2\sqrt{3})^2 + (y - 1)^2 = 1$ (b) $(x - 2\sqrt{3})^2 + \left(y + \frac{1}{2}\right)^2 = 1$
 (c) $(x - \sqrt{3})^2 + (y + 1)^2 = 1$ (d) $(x - \sqrt{3})^2 + (y - 1)^2 = 1$

Q 27. Points E and F are given by

- (a) $\left(\frac{\sqrt{3}}{2}, \frac{3}{2}\right), (\sqrt{3}, 0)$ (b) $\left(\frac{\sqrt{3}}{2}, \frac{1}{2}\right), (\sqrt{3}, 0)$ (c) $\left(\frac{\sqrt{3}}{2}, \frac{3}{2}\right), \left(\frac{\sqrt{3}}{2}, \frac{1}{2}\right)$ (d) $\left(\frac{3}{2}, \frac{\sqrt{3}}{2}\right), \left(\frac{\sqrt{3}}{2}, \frac{1}{2}\right)$

Q 28. Equations of the sides QR, RP are

- (a) $y = \frac{2}{\sqrt{3}} + 1, y = -\frac{2}{\sqrt{3}}x - 1$ (b) $y = \frac{x}{\sqrt{3}}, y = 0$
 (c) $y = \frac{\sqrt{3}}{2}x + 1, y = -\frac{\sqrt{3}}{2}x - 1$ (d) $y = \sqrt{3}x, y = 0$

Paragraph for Question Nos. 29 to 31

Consider the lines: $L_1: \frac{x+1}{3} = \frac{y+2}{1} = \frac{z+1}{2}$
 $L_2: \frac{x-2}{1} = \frac{y+2}{2} = \frac{z-3}{3}$

Q 29. The unit vector perpendicular to both L_1 and L_2 is

- (a) $\frac{1}{\sqrt{99}}(i + 7j + 7k)$ (b) $\frac{1}{5\sqrt{3}}(-i - 7j + 7k)$ (c) $\frac{1}{5\sqrt{3}}(-i + 7j + 5k)$ (d) $\frac{1}{\sqrt{99}}(7i - 7j - k)$

Q 30. The shortest distance between L_1 and L_2 is

- (a) 0 (b) $\frac{17}{\sqrt{3}}$ (c) $\frac{41}{5\sqrt{3}}$ (d) $\frac{17}{5\sqrt{3}}$

Q 31. The distance of the point (1, 1, 1) from the plane passing through the point (-1, -2, -1) and whose normal is perpendicular to both the lines, L_1 and L_2 is

- (a) $\frac{2}{\sqrt{75}}$ (b) $\frac{7}{\sqrt{75}}$ (c) $\frac{13}{\sqrt{75}}$ (d) $\frac{23}{\sqrt{75}}$

Paragraph for Question Nos. 32 to 34

Consider the functions defined implicitly by the equation $y - 3y + x = 0$ on various intervals in the real line.

If $x \in (-\infty, -2) \cup (2, \infty)$, the equation implicitly defines a unique real valued differentiable function $y = f(x)$.

If $x \in (-2, 2)$, the equation implicitly defines a unique real valued differentiable function $y = g(x)$ satisfying $g(0) = 0$.

Q 32. If $f(-10\sqrt{2}) = 2 > \sqrt{2}$ then $f'(-10\sqrt{2}) =$

- (a) $\frac{4\sqrt{2}}{7^3 \cdot 3^2}$ (b) $-\frac{4\sqrt{2}}{7^3 \cdot 3^2}$ (c) $\frac{4\sqrt{2}}{7^3 \cdot 3}$ (d) $-\frac{4\sqrt{2}}{7^3 \cdot 3}$

- Q 33. The area of the region bounded by the curve $y = f(x)$, x -axis, and the lines $x = a$ and $x = b$, where $-\infty < a < b < -2$, is

$$\begin{aligned} \text{(a)} \quad & \int_a^b \frac{x}{3\{(f(x))^2 - 1\}} dx + bf(b) - af(a) & \text{(b)} \quad & -\int_a^b \frac{x}{3\{(f(x))^2 - 1\}} dx + bf(b) - af(a) \\ \text{(c)} \quad & \int_a^b \frac{x}{3\{(f(x))^2 - 1\}} dx - bf(b) + af(a) & \text{(d)} \quad & -\int_a^b \frac{x}{3\{(f(x))^2 - 1\}} dx - bf(b) + af(a) \end{aligned}$$

Q 34. $\int_{-1}^1 g'(x) dx =$

- (a) $2g(-1)$ (b) 0 (c) $-2g(1)$ (d) $2g(1)$

Assertion-Reasoning Type

- Q 35. Suppose four distinct positive numbers a_1, a_2, a_3, a_4 are in GP. Let

$$b_1 = a_1, b_2 = b_1 + a_2, b_3 = b_2 + b_3 \text{ and } b_4 = b_3 + a_4.$$

STATEMENT-1: The numbers b_1, b_2, b_3, b_4 are neither in AP nor in GP, and

STATEMENT-2: The numbers b_1, b_2, b_3, b_4 are in HP.

(a) STATEMENT-1 is True, STATEMENT-2 is True; STATEMENT-2 is a correct explanation for STATEMENT-1.

(b) STATEMENT-1 is True, STATEMENT-2 is True; STATEMENT-2 is NOT a correct explanation for STATEMENT-1.

(c) STATEMENT-1 is True, STATEMENT-2 is False.

(d) STATEMENT-1 is False, STATEMENT-2 is True.

- Q 36. Consider the three planes $P_1: x - y + z = 1$

$$P_2: x + y - z = -1$$

$$P_3: x - 3y + 3z = 2$$

Let L_1, L_2, L_3 be the lines of intersection of the planes P_2 and P_3 , P_3 and P_1 , and P_1 and P_2 respectively.

STATEMENT-1: At least two of the lines L_1, L_2 and L_3 are non-parallel, and

STATEMENT-2: The three planes do not have a common point.

(a) STATEMENT-1 is True, STATEMENT-2 is True;

STATEMENT-2 is a correct explanation for STATEMENT-1.

(b) STATEMENT-1 is True, STATEMENT-2 is True;

STATEMENT-2 is NOT a correct explanation for STATEMENT-1.

(c) STATEMENT-1 is True, STATEMENT-2 is False.

(d) STATEMENT-1 is False, STATEMENT-2 is True.

- Q 37. Consider the system of equations

$$x - 2y + 3z = -1$$

$$-x + y - 2z = k$$

$$x - 3y + 4z = 1$$

STATEMENT-1: The system of equations has no solution for $k \neq 3$, and

STATEMENT-2: The determinant $\begin{vmatrix} 1 & 3 & -1 \\ -1 & -2 & k \\ 1 & 4 & 1 \end{vmatrix} \neq 0$, for $k \neq 3$.

(a) STATEMENT-1 is True, STATEMENT-2 is True;

STATEMENT-2 is a correct explanation for STATEMENT-1.

(b) STATEMENT-1 is True, STATEMENT-2 is True;

STATEMENT-2 is NOT a correct explanation for STATEMENT-1.

(c) STATEMENT-1 is True, STATEMENT-2 is False.

(d) STATEMENT-1 is False, STATEMENT-2 is True.

- Q 38. Consider the system of equations

$$ax + by = 0, cx + dy = 0; \text{ where } a, b, c, d \in \{0, 1\}.$$

STATEMENT-1: The probability that the system of equations has a unique solution is $\frac{3}{8}$, and

STATEMENT-2: The probability that the system of equations has a solution is 1.

(a) STATEMENT-1 is True, STATEMENT-2 is True; STATEMENT-2 is a correct explanation for STATEMENT-1.

- (b) STATEMENT-1 is True, STATEMENT-2 is True; STATEMENT-2 is NOT a correct explanation for STATEMENT-1.
 (c) STATEMENT-1 is True, STATEMENT-2 is False.
 (d) STATEMENT-1 is False, STATEMENT-2 is True.
- Q 39. Let f and g be real-valued functions defined on interval $(-1, 1)$ such that $g''(x)$ is continuous, $g(0) \neq 0$, $g'(0) = 0$, $g''(0) \neq 0$ and $f(x) = g(x) \cdot \sin x$.
 STATEMENT-1: $\lim_{x \rightarrow 0} \{g(x) \cdot \cot x - g(0) \cdot \operatorname{cosec} x\} = f'(0)$, and
 STATEMENT-2: $f(0) = g(0)$.
 (a) STATEMENT-1 is True, STATEMENT-2 is True; STATEMENT-2 is a correct explanation for STATEMENT-1.
 (b) STATEMENT-1 is True, STATEMENT-2 is True; STATEMENT-2 is NOT a correct explanation for STATEMENT-1.
 (c) STATEMENT-1 is True, STATEMENT-2 is False.
 (d) STATEMENT-1 is False, STATEMENT-2 is True.
- Q 40. Consider $L_1: 2x + 3y + p - 3 = 0$
 $L_2: 2x + 3y + p + 3 = 0$.
 where p is a real number and $C: x^2 + y^2 + 6x - 10y + 30 = 0$.
 STATEMENT-1: If the line L_1 is a chord of the circle C then the line L_2 is not always a diameter of the circle C and
 STATEMENT-2: If the line t_1 is a diameter of the circle C then the line L_2 is not a chord of the circle C .
 (a) STATEMENT-1 is True, STATEMENT-2 is True; STATEMENT-2 is a correct explanation for STATEMENT-1.
 (b) STATEMENT-1 is True, STATEMENT-2 is True; STATEMENT 2 is NOT a correct explanation for STATEMENT-1.
 (c) STATEMENT-1 is True, STATEMENT-2 is False.
 (d) STATEMENT-1 is False, STATEMENT-2 is True.
- Q 41. Let a, b, c, p, q be real numbers. Suppose α, β are the roots of the equation $x^2 + 2px + q = 0$ and $\alpha, \frac{1}{\beta}$ are the roots of the equation $ax^2 + 2bx + c = 0$ where $\beta^2 \notin \{-1, 0, 1\}$.
 STATEMENT-1: $(p^2 - q)(b^2 - ac) \geq 0$, and
 STATEMENT-2: $b \neq pa$ or $c \neq qa$
 (a) STATEMENT-1 is True, STATEMENT-2 is True; STATEMENT-2 is a correct explanation for STATEMENT-1.
 (b) STATEMENT-1 is True, STATEMENT-2 is True; STATEMENT-2 is NOT a correct explanation for STATEMENT-1.
 (c) STATEMENT-1 is True, STATEMENT-2 is False.
 (d) STATEMENT-1 is False, STATEMENT-2 is True.
- Q 42. Let a solution $y = y(x)$ of the differential equation

$$x\sqrt{x^2 - 1}dy - y\sqrt{y^2 - 1}dx = 0$$

 satisfy $y(2) = \frac{2}{\sqrt{3}}$.
 STATEMENT-1: $y(x) = \sec\left(\sec^{-1} x - \frac{\pi}{6}\right)$, and
 STATEMENT-2: $y(x)$ is given by $\frac{1}{y} = \frac{2\sqrt{3}}{x} - \sqrt{1 - \frac{1}{x^2}}$
 (a) STATEMENT-1 is True, STATEMENT-2 is True; STATEMENT-2 is a correct explanation for STATEMENT-1.
 (b) STATEMENT-1 is True, STATEMENT-2 is True; STATEMENT-2 is NOT a correct explanation for STATEMENT-1.
 (c) STATEMENT-1 is True, STATEMENT-2 is False.
 (d) STATEMENT-1 is False, STATEMENT-2 is True.

Matrix Match Type Questions

Statements A, B, C, D in Column I have to be matched with statements p, q, r, s in Column II. The answers to these questions have to be appropriately bubbled as illustrated below.

If the correct matches are A-q, A-r, B-p, B-s, C-r, C-s, D-q then the correctly bubbled matrix will be the following.

	p	q	r	s
A	☐	☒	☒	☐
B	☒	☐	☐	☒
C	☐	☐	☒	☒
D	☐	☒	☐	☐

- Q 43. Consider the lines given by $L_1: x + 3y - 5 = 0$
 $L_2: 3x - ky - 1 = 0$ $L_3: 5x + 2y - 12 = 0$.

Column I

- (A) L_1, L_2, L_3 are concurrent, if
 (B) One of L_1, L_2, L_3 is parallel to at least one
 (C) L_1, L_2, L_3 form a triangle, if
 (D) L_1, L_2, L_3 do not form a triangle, if

Column II

- (p) $k = -9$
 (q) $k = -$ of the other two, if
 (r) $k = \frac{5}{6}$
 (s) $k = 5$

- Q 44. Consider all possible permutations of letters of the word ENDEANOEL.

Column I

- (A) The number of permutations containing the
 (B) The number of permutations in which the
 (C) The number of permutations in which none
 (D) The number of permutations in which the

Column II

- (p) [5 word ENDEA is
 (q) 2 [5 letter E occurs in the first and the last positions is
 (r) 7 of the letters D, L, N occurs in the last five positions
 (s) 21 letters A, E, O occur only in odd positions is

- Q 45. **Column I**

- (A) The minimum positive value of $\frac{x^2 + 2x + 4}{x + 2}, x \in \mathbb{R}$ is
 (B) Let A and B be 3×3 matrices of real numbers, where A is symmetric, B is skew-symmetric and $(A + B)(A - B) = (A - B)(A + B)$. If $(AB)^T = (-1)^k AB$, where $(AB)^T$ is the transpose of AB then the possible values of k are
 (C) Let $a = \log_3 \log_3 2$. An integer k satisfying

Column II

- (p) 0

- $1 < 2^{-k+3^{-a}} < 2$, must be less than
 (D) If $\sin \theta = \cos \phi$, then the possible values of

- (q) 1

- (r) 2

- (s) 3

- $\frac{1}{\pi} \left(\theta \pm \phi - \frac{\pi}{2} \right)$ are

Answers

1c	2c	3c	4a	5c	6b	7b	8b	9a	10d
11c	12a	13d	14d	15b	16a,b,c,d	17a, d	18b,d	19b,c	20b
21c	22d	23 a	24a	25b	26d	27a	28d	29b	30d
31c	32b	33a	34d	35c	36d	37a	38b	39b	40c
41b	42c								

43.

	p	q	r	s
A	☐	☐	☐	☒
B	☒	☒	☐	☐
C	☐	☐	☒	☐
D	☒	☒	☐	☒

44.

	p	q	r	s
A	☒	☐	☐	☐
B	☐	☐	☐	☒
C	☐	☒	☐	☐
D	☐	☒	☐	☐

45.

	p	q	r	s
A	☐	☐	☒	☐
B	☐	☒	☐	☒
C	☐	☐	☒	☒
D	☒	☐	☒	☐

IIT Questions

SECTION-I

Single-Correct Choice Type

Each question in this section has 4 choices (a), (b), (c) and (d) for its answer, out of which ONLY ONE is correct.

- Q 1. If the sum of the first n terms of an AP is cn^2 then the sum of squares of these n terms is
 (a) $\frac{n(4n^2 - 1)c^2}{6}$ (b) $\frac{n(4n^2 + 1)c^2}{3}$ (c) $\frac{n(4n^2 - 1)c^2}{3}$ (d) $\frac{n(4n^2 + 1)c^2}{6}$
- Q 2. Let $z = x + iy$ be a complex number where x and y are integers. Then the area of the rectangle whose vertices are the roots of the equation $zz^3 + \bar{z}\bar{z}^3 = 350$ is
 (a) 48 (b) 32 (c) 40 (d) 80
- Q 3. Let $z = \cos \theta + i \sin \theta$. Then the value of $\sum_{m=1}^{15} \text{Im}(z^{2m-1})$ at $\theta = 2^\circ$ is
 (a) $\frac{1}{\sin 2^\circ}$ (b) $\frac{1}{3 \sin 2^\circ}$ (c) $\frac{1}{2 \sin 2^\circ}$ (d) $\frac{1}{4 \sin 2^\circ}$
- Q 4. The number of seven digit integers, with sum of the digits equal to 10 and formed by using the digits 1, 2 and 3 only, is
 (a) 55 (b) 66 (c) 77 (d) 88
- Q 5. The locus of orthocentre of the triangle formed by the lines $(1+p)x - py + p(1+p) = 0$, $(1+q)x - qy + q(1+q) = 0$ and $y = 0$ where $p \neq q$, is
 (a) a hyperbola (b) a parabola (c) an ellipse (d) a straight line
- Q 6. Tangents drawn from the point $P(1, 8)$ to the circle $x^2 + y^2 - 6x - 4y - 11 = 0$ touch the circle at the points A and B. The equation of the circumcircle of the $\triangle PAB$ is
 (a) $x^2 + y^2 + 4x - 6y + 19 = 0$ (b) $x^2 + y^2 - 4x - 10y + 19 = 0$
 (c) $x^2 + y^2 - 2x + 6y - 29 = 0$ (d) $x^2 + y^2 - 6x - 4y + 19 = 0$
- Q 7. The line passing through the extremity A of the major axis and extremity B of the minor axis of the ellipse $x^2 + 9y^2 = 9$ meets its auxiliary circle at the point M. Then the area of the triangle with vertices at A, M and the origin O is
 (a) $\frac{31}{10}$ (b) $\frac{29}{10}$ (c) $\frac{21}{10}$ (d) $\frac{27}{10}$
- Q 8. The normal at a point P on the ellipse $x^2 + 4y^2 = 16$ meets the x-axis at Q. If M is the mid-point of the line segment PQ then the locus of M intersects the latus rectum of the given ellipse at the points
 (a) $\left(\pm \frac{3\sqrt{5}}{2}, \pm \frac{2}{7}\right)$ (b) $\left(\pm \frac{3\sqrt{5}}{2}, \pm \frac{\sqrt{19}}{4}\right)$ (c) $\left(\pm \sqrt{3}, \pm \frac{1}{7}\right)$ (d) $\left(\pm 2\sqrt{3}, \pm \frac{4\sqrt{3}}{7}\right)$
- Q 9. If $\vec{a}, \vec{b}, \vec{c}$ and $\vec{d}$ are unit vectors such that $(\vec{a} \times \vec{b}) \cdot (\vec{c} \times \vec{d}) = 1$ and $\vec{a} \cdot \vec{c} = \frac{1}{2}$ then
 (a) $\vec{a}, \vec{b}, \vec{c}$ are non-coplanar (b) $\vec{b}, \vec{c}, \vec{d}$ are non-coplanar
 (c) $\vec{b}, \vec{d}$ are non-parallel (d) $\vec{a}, \vec{d}$ are parallel and $\vec{b}, \vec{c}$ are parallel
- Q 10. Let $P(3, 2, 6)$ be a point in space and Q be a point on the line $\vec{r} = (i - j + 2k) + \mu(-3i + j + 5k)$. Then the value of μ for which the vector $\vec{PQ}$ is parallel to the plane $x - 4y + 3z = 1$ is
 (a) $\frac{1}{4}$ (b) $-\frac{1}{4}$ (c) $\frac{1}{8}$ (d) $-\frac{1}{8}$
- Q 11. A line with positive direction cosines passes through the point $P(2, -1, 2)$ and makes equal angles with the coordinate axes. The line meets the plane $2x + y + z = 9$ at point Q. The length of the line segment PQ equals
 (a) 1 (b) $\sqrt{2}$ (c) $\sqrt{3}$ (d) 2

- Q 12. Let f be a non-negative function defined on the interval $[0, 1]$. If $\int_0^x \sqrt{1 - \{f'(t)\}^2} dt = \int_0^x f(t) dt, 0 \leq x \leq 1$
 $f(0) = 0$ then
- (a) $f\left(\frac{1}{2}\right) < \frac{1}{2}$ and $f\left(\frac{1}{3}\right) > \frac{1}{3}$ (b) $f\left(\frac{1}{2}\right) > \frac{1}{2}$ and $f\left(\frac{1}{3}\right) > \frac{1}{3}$
(c) $f\left(\frac{1}{2}\right) < \frac{1}{2}$ and $f\left(\frac{1}{3}\right) < \frac{1}{3}$ (d) $f\left(\frac{1}{2}\right) > \frac{1}{2}$ and $f\left(\frac{1}{3}\right) < \frac{1}{3}$

SECTION II

Multiple Correct Choice Type

Each question in this section has four choices (a), (b), (c) and (d) for its answer, out of which ONE OR MORE is/are correct.

- Q 13. If $\frac{\sin^4 x}{2} + \frac{\cos^4 x}{3} = \frac{1}{5}$, then
- (a) $\tanh 2x = \frac{3}{2}$ (b) $\frac{\sin^8 x}{8} + \frac{\cos^8 x}{27} = \frac{1}{125}$ (c) $\tan^2 x = \frac{2}{3}$ (d) $\frac{\sin^8 x}{8} + \frac{\cos^8 x}{27} = \frac{2}{125}$
- Q 14. For $0 < \theta < \frac{\pi}{2}$, the solution(s) of
- $$\sum_{m=1}^6 \operatorname{cosec}\left(\theta + \frac{m-1}{4}\pi\right) \cdot \operatorname{cosec}\left(\theta + \frac{m}{4}\pi\right) = 4\sqrt{2} \text{ is/are}$$
- (a) $\frac{\pi}{4}$ (b) $\frac{\pi}{6}$ (c) $\frac{\pi}{12}$ (d) $\frac{5\pi}{12}$
- Q 15. In a triangle ABC with fixed base BC, the vertex A moves such that $\cos B + \cos C = 4\sin^2 \frac{A}{2}$. If a , b and c denote the lengths of the sides of the triangle opposite to the angles A, B and C respectively then
- (a) $b + c = 4a$ (b) $b + c = 2a$
(c) locus of A is an ellipse (d) locus of A is a pair of straight lines
- Q 16. The tangent PT and the normal PN to the parabola $y^2 = 4ax$ at a point P on it meet its axis at points T and N respectively. The locus of the centroid of the Δ PTN is a parabola whose
- (a) vertex = $\left(\frac{2a}{3}, 0\right)$ (b) directrix is $x = 0$ (c) latus rectum = $\frac{2a}{3}$ (d) focus = $(a, 0)$
- Q 17. An ellipse intersects the hyperbola $2x^2 - 2y^2 = 1$ orthogonally. The eccentricity of the ellipse is reciprocal of that of the hyperbola. If the axes of the ellipse are along the coordinate axes, then the
- (a) equation of the ellipse is $x^2 + 2y^2 = 2$ (b) foci of the ellipse are $(\pm 1, 0)$
(c) equation of the ellipse is $x^2 + 2y^2 = 4$ (d) foci of the ellipse are $(+\sqrt{2}, 0)$
- Q 18. For the function $f(x) = x \cos \frac{1}{x}, x \geq 1$
- (a) for at least one $x \in [1, \infty)$, $f(x+2) - f(x) < 2$ (b) $\lim_{x \rightarrow \infty} f'(x) = 1$
(c) for all $x \in [1, \infty)$, $f(x+2) - f(x) > 2$ (d) $f'(x)$ is strictly decreasing in $[1, \infty)$
- Q 19. If $I_n = \int_{-\pi}^{\pi} \frac{\sin nx}{(1 + \pi^x) \cdot \sin x} dx, n=0, 1, 2, \dots$, then
- (a) $I_n = I_{n+2}$ (b) $\sum_{m=1}^{10} I_{2m+1} = 10\pi$ (c) $\sum_{m=1}^{10} I_{2m+1} = 0$ (d) $I_n = I_{n+1}$
- Q 20. The area of the region bounded by the curve $y = e^x$ and lines $x = 0$ and $y = e$ is
- (a) $e - 1$ (b) $\int_1^e \log_e(e+1-y) dy$ (c) $e - \int_0^1 e^x dx$ (d) $\int_1^e \log_e y dy$

- Q 21. Let $L = \lim_{x \rightarrow 0} \frac{a - \sqrt{a^2 - x^2} - \frac{x^2}{4}}{x^4}$, $a > 0$. If L is finite, then
- (a) $a = 2$ (b) $a = 1$ (c) $L = \frac{1}{64}$ (d) $L = \frac{1}{32}$

SECTION III

Comprehension Type

In this section, each group has 3 multiple choice questions based on a paragraph. Each question has 4 choices (a), (b), (c) and (d) for its answer, out of which **ONLY ONE** is correct.

Paragraph for Q. N. 22 to 24

Let A be the set of all 3×3 symmetric matrices all of whose elements are either 0 or 1. Five of the elements are 1 and four are 0.

- Q 22. The number of matrices in A is
 (a) 12 (b) 6 (c) 9 (d) 3
- Q 23. The number of matrices A in A for which the system of linear equations $A \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ has a unique solution, is
 (a) less than 4 (b) at least 4 but less than 7
 (c) at least 7 but less than 10 (d) at least 10

- Q 24. The number of matrices A in A for which the system of linear equations $A \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ is inconsistent, is
 (a) 0 (b) more than 2 (c) 2 (d) 1

Paragraph for Q. N. 25 to 27

A fair die is tossed repeatedly until a six is obtained. Let X denote the number of tosses required.

- Q 25. The probability that $X = 3$ equals
 (a) $\frac{25}{216}$ (b) $\frac{25}{36}$ (c) $\frac{5}{36}$ (d) $\frac{125}{216}$
- Q 26. The probability that $X > 3$ equals
 (a) $\frac{125}{126}$ (b) $\frac{25}{216}$ (c) $\frac{5}{36}$ (d) $\frac{25}{36}$
- Q 27. The conditional probability that $X \geq 6$ given $X > 3$ equals
 (a) $\frac{125}{216}$ (b) $\frac{25}{216}$ (c) $\frac{5}{36}$ (d) $\frac{25}{36}$

SECTION IV

Matrix-Match Type

In this section each question contains statements given in two columns, which are to be matched. The statements in Column I are labelled (A), (B), (C) and (D), while the statements in Column II are labelled (p), (q), (r), (s) and (t). Any given statement in Column I can have correct matching with **ONE OR MORE** statement(s) in Column II. The appropriate bubbles corresponding to the answers to these questions have to be darkened as illustrated in the following example.

If the correct matches are A—p, s and t; B—q and r; C—p and q; D—s then the correct darkening of bubbles will look like the following.

	p	q	r	s	t
A	☒	☐	☐	☒	☒
B	☐	☒	☒	☐	☐
C	☒	☒	☐	☐	☐
D	☐	☐	☐	☒	☐

Q 28. Match the statements/expressions given in Column I with the values given in Column II.

Column I	Column II
(A) Root(s) of the equation $2\sin^2\theta + \sin^2 2\theta = 2$	(p) $\pi/6$
(B) Points of discontinuity of $f(x) = \left[\frac{6x}{\pi} \right] \cos \left[\frac{3x}{\pi} \right]$ where $[y]$ = greatest integer less than or equal to y	(q) $\pi/4$ (r) $\pi/3$
(C) The volume of the parallelopiped with edges represented by the vectors $\vec{i} + \vec{j}, \vec{i} + 2\vec{j}$ and $\vec{i} + \vec{j} + \pi\vec{k}$	(s) $\pi/2$
(D) The angle between vectors $\vec{a}$ and $\vec{b}$ where $\vec{a}, \vec{b}$ and $\vec{c}$ are unit vectors satisfying- $\vec{a} + \vec{b} + \sqrt{3}\vec{c} = \vec{0}$	(t) π

Q 29. Match the statements/expressions given in Column I with the values given in Column II.

Column I	Column II
(A) The number of solutions of the equation $x^{e^{\sin x}} - \cos x = 0$ in the interval $\left(0, \frac{\pi}{2}\right)$	(P) 1 (q) 2
(B) The value(s) of k for which the planes $kx + 4y + z = 0$, $4x + ky + 2z = 0$ and $2x + 2y + z = 0$ intersect in a straight line	(r) 3
(C) the value(s) of k for which $ x - 1 + x - 2 + x + 1 + x + 2 = 4k$ has integral solution(s)	(s) 4
(D) If $y' = y + 1$ and $y(0) = 1$ then value(s) of $y(\log_e 2)$	(t) 5

Q 30. Match the statements/expressions in Column I with the open intervals in Column II.

Column I	Column II
(A) The interval contained in the domain of definition of non-zero solutions of the differential equation $(x - 3)^2 \cdot y' + y = 0$	(P) $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$
(B) The interval containing the value of the integral $\int_1^5 (x - 1)(x - 2)(x - 3)(x - 4)(x - 5) dx$	(q) $\left(0, \frac{\pi}{8}\right)$ (r) $\left(\frac{\pi}{8}, \frac{5\pi}{4}\right)$
(C) The interval in which at least one of the points of local maximum of $\cos x + \sin x$ lies	(s) $\left(0, \frac{\pi}{8}\right)$

(D) The interval in which $\tan^{-1}(\sin x + \cos x)$ is increasing	(t) $(-\pi, \pi)$
--	-------------------

Q 31. Match the conic sections in Column I with the statements/expressions in Column II.

Column I	Column II
(A) Circle	(p) The locus of the point (h, k) for which the line $hx + ky = 1$ touches the circle $x^2 + y^2 = 4$ (q) Points z in the complex plane satisfying $ z + 2 - z - 2 = \pm 3$
(B) Parabola	(r) Points of the conic have parametric representation $x = \sqrt{3} \left(\frac{1-t^2}{1+t^2} \right), y = \frac{2t}{1+t^2}$
(C) Ellipse	(s) The eccentricity of the conic lies in the interval $1 \leq e \leq \infty$
(D) Hyperbola	(t) Points z in the complex plane satisfying $\operatorname{Re}(z + 1)^2 = z ^2 + 1$

SECTION V

Integer Answer Type

In this section the answer to each of the questions is a single-digit integer, ranging from 0 to 9. If the correct answers to question numbers X, Y, Z and W (say) are 6, 0, 9 and 2 respectively, then the correct darkening of bubbles will look like the following:

X	Y	Z	W
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

- Q 32. The centres of two circles C_1 and C_2 each of unit radius are at a distance of 6 units from each other. Let P be the mid-point of the line segment joining the centres of C_1 and C_2 and C be a circle touching circles C_1 and C_2 externally. If a common tangent to C_1 and C passing through P is also a common tangent to C_2 and C then the radius of the circle C is
- Q 33. Let ABC and ABC' be two non-congruent triangles with sides $AB = 4$, $AC = AC' = 2\sqrt{2}$ and angle $B = 30^\circ$. The absolute value of the difference between the areas of these triangles is
- Q 34. Let (x, y, z) be points with integral coordinates satisfying the system of homogeneous equations $3x - y - z = 0$, $-3x + z = 0$, $-3x + 2y + z = 0$. Then the number of such points for which $x^2 + y^2 + z^2 \leq 100$ is
- Q 35. The smallest value of k for which both the roots of the equation $x^2 - 8kx + 16(k^2 - k + 1) = 0$ are real, distinct and have values at least 4, is
- Q 36. The maximum value of the function $f(x) = 2x^3 - 15x^2 + 36x - 48$ on the set $A = \{x | x^2 + 20 < 9x\}$ is
- Q 37. If the function $f(x) = x^3 + e^{x/2}$ and $g(x) = f^{-1}(x)$ then the value of $g'(1)$ is

IIT Questions

SECTION-I

Single-Correct Choice Type

This section contains 14 multiple-choice questions. Each question has 4 choices (a), (b), (c) and (d) for its answer, out of which only one is correct.

- Q 1. Let p and q be real numbers such that $p \neq 0$, $p^3 \neq q$ and $p \neq -q$. If α and β are nonzero complex numbers satisfying $\alpha + \beta = -p$ and $\alpha^3 + \beta^3 = q$ then a quadratic equation having $\frac{\alpha}{\beta}$ and $\frac{\beta}{\alpha}$ as its roots is
- (a) $(p^3 + q)x^2 - (p^3 + 2q)x + (p^3 + q) = 0$ (b) $(p^3 + q)x^2 - (p^3 - 2q)x + (p^3 + q) = 0$
 (c) $(p^3 - q)x^2 - (5p^3 - 2q)x + (p^3 - q) = 0$ (d) $(p^3 - q)x^2 - (5p^3 + 2q)x + (p^3 - q) = 0$
- Q 2. Let $S = \{1, 2, 3, 4\}$. The total number of unordered pairs of disjoint subsets of S is equal to
 (a) 25 (b) 34 (c) 42 (d) 41
- Q 3. For $r = 0, 1, 2, 3, \dots, 10$ let A_r, B_r, C_r denote respectively the coefficient of x^r in the expansions of $(1 + x)^{10}$, $(1 + x)^{20}$ and $(1 + x)^{30}$. Then $\sum_{r=1}^{10} A_r (B_{10}B_r - C_{10}A_r)$ is equal to
 (a) $B_{10} - C_{10}$ (b) $A_{10}(B_{10}^2 - C_{10}A_{10})$ (c) 0 (d) $C_{10} - B_{10}$
- Q 4. The number of 3×3 matrices A whose entries are either 0 or 1 and for which the system $A \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ has exactly two distinct solutions, is
 (a) 0 (b) $2^9 - 1$ (c) 168 (d) 2
- Q 5. If the angles A, B and C of a triangle are in an AP and if a, b and c denote the lengths of the sides opposite to A, B and C respectively then the value of the expression $\frac{a}{c} \sin 2C + \frac{c}{a} \sin 2A$ is
 (a) $\frac{1}{2}$ (b) $\frac{\sqrt{3}}{2}$ (c) 1 (d) $\sqrt{3}$
- Q 6. Let P, Q, R and S be points on the plane with position vectors $-2\mathbf{i} - \mathbf{j}$, $4\mathbf{i}$, $3\mathbf{i} + 3\mathbf{j}$ and $-3\mathbf{i} + 2\mathbf{j}$ respectively. The quadrilateral PQRS must be a
 (a) parallelogram, which is neither a rhombus nor a rectangle
 (b) square
 (c) rectangle but not a square
 (d) rhombus but not a square
- Q 7. Two adjacent sides of a parallelogram ABCD are given by $\vec{AB} = 2\mathbf{i} + 10\mathbf{j} + 11\mathbf{k}$ and $\vec{AD} = \mathbf{i} + 2\mathbf{j} + 2\mathbf{k}$. The side AD is rotated by an acute angle α in the plane of the parallelogram so that AD becomes AD'. If AD' makes a right angle with the side AB then the cosine of the angle α is given by
 (a) $\frac{8}{9}$ (b) $\frac{\sqrt{17}}{9}$ (c) $\frac{1}{9}$ (d) $\frac{4\sqrt{5}}{9}$
- Q 8. The equation of the plane containing the straight line $\frac{x}{2} = \frac{y}{3} = \frac{z}{4}$ and perpendicular to the plane containing the straight lines $\frac{x}{3} = \frac{y}{4} = \frac{z}{2}$ and $\frac{x}{4} = \frac{y}{2} = \frac{z}{3}$ is
 (a) $x + 2y - 2z = 0$ (b) $3x + 2y - 2z = 0$ (c) $x - 2y + z = 0$ (d) $5x + 2y - 4z = 0$
- Q 9. If the distance of the point $P(1, -2, 1)$ from the plane $x + 2y - 2z = \alpha$, where $\alpha > 0$, is 5 then the foot of the perpendicular from P to the plane is
 (a) $\left(\frac{8}{3}, \frac{4}{3}, \frac{-7}{3}\right)$ (b) $\left(\frac{4}{3}, -\frac{4}{3}, \frac{1}{3}\right)$ (c) $\left(\frac{1}{3}, \frac{2}{3}, \frac{10}{3}\right)$ (d) $\left(\frac{2}{3}, -\frac{1}{3}, \frac{5}{2}\right)$

- Q 10. Let f, g and h be real-valued functions defined on the interval $[0, 1]$ by $f(x) = e^{x^2} + -e^{-x^2}$, $g(x) = xe^{-x^2}$ and $hx = x^2e^{x^2} + e^{-x^2}$. If a, b and c denote the absolute maximum of f, g and h on $[0, 1]$ then
 (a) $a = b$ and $b \neq c$ (b) $c = a$ and $a \neq b$ (c) $a \neq b$ and $b \neq c$ (d) $a = b = c$
- Q 11. The value of $\lim_{x \rightarrow 0} \frac{1}{x^3} \int_0^x \frac{t \log(1+t)}{t^4 + 4} dt$ is
 (a) 0 (b) $\frac{1}{12}$ (c) $\frac{1}{24}$ (d) $\frac{1}{64}$
- Q 12. Let f be a real-valued function defined on the interval $(-1, 1)$ such that $e^{-x} \cdot f(x) = 2 + \int_0^x \sqrt{t^4 + 1} dt$
 $\forall x \in (-1, 1)$. Let f^{-1} be the inverse function of f . Then $(f^{-1})'(2)$ is equal to
 (a) 1 (b) $\frac{1}{3}$ (c) $\frac{1}{2}$ (d) $\frac{1}{e}$
- Q 13. Let ω be a complex cube root of unity with $\omega \neq 1$. A fair die is thrown three times. If r_1, r_2 and r_3 are the numbers obtained on the die then the probability that $\omega^{r_1} + \omega^{r_2} + \omega^{r_3} = 0$ is
 (a) $\frac{1}{18}$ (b) $\frac{1}{9}$ (c) $\frac{2}{9}$ (d) $\frac{1}{26}$
- Q 14. A signal which can be green or red with probability $\frac{4}{5}$ and $\frac{1}{5}$ respectively is received by station A and then transmitted to station B. The probability of each station receiving the signal correctly is $\frac{3}{4}$. If the signal received at station B is green then the probability that the original signal was green is
 (a) 3 (b) $\frac{6}{7}$ (c) $\frac{20}{23}$ (d) $\frac{9}{20}$

SECTION II

Multiple Correct Choice Type

This section contains 5 multiple-choice questions. Each question has four choices (a), (b), (c) and (d) out of which one or more may be correct.

- Q 15. Let z_1 and z_2 be two distinct complex numbers and let $z = (1 - t)z_1 + tz_2$ for some real number t , where $0 < t < 1$. If $\arg(w)$ denotes the principal argument of a nonzero complex number w then
 (a) $|z - z_1| + |z - z_2| = |z_1 - z_2|$
 (b) $\arg(z - z_1) = \arg(z - z_2)$
 (c) $\left| \frac{z - z_1}{z_2 - z_1} \cdot \frac{\bar{z} - \bar{z}_1}{\bar{z}_2 - \bar{z}_1} \right| = 0$
 (d) $\arg(z - z_1) = \arg(z_2 - z_1)$
- Q 16. Let ABC be a triangle such that $\angle ACB = \frac{\pi}{6}$ and let a, b, c denote the lengths of the sides opposite A, B, C respectively. The value(s) of x for which $a = x^2 + x + 1$, $b = x^2 - 1$ and $c = 2x + 1$ is (are)
 (a) $-2 + \sqrt{3}$ (b) $\sqrt{3} + 1$ (c) $2 + \sqrt{3}$ (d) $4 + \sqrt{3}$
- Q 17. Let A and B be two distinct points on the parabola $y = 4x$. If the axis of the parabola touches a circle of radius r having AB as its diameter then the slope of the line AB can be
 (a) $-\frac{1}{r}$ (b) $\frac{1}{r}$ (c) $\frac{2}{r}$ (d) $-\frac{2}{r}$

- Q 18. The value(s) of $\int_0^1 \frac{x^4(1-x)^4}{1+x^2} dx$ is (are)
- (a) $\frac{22}{7} - \pi$ (b) $\frac{2}{105}$ (c) 0 (d) $\frac{71}{15} - \frac{3\pi}{2}$
- Q 19. Let f be a real-valued function defined on the interval $(0, \infty)$ by $f(x) = \log_e x + At$. Which of the following statement(s) is (are) true?
- (a) $f''(x)$ exists for all $x \in (0, \infty)$
 (b) $f'(x)$ exists for all $x \in (0, \infty)$ and $f'(x)$ is continuous on $(0, \infty)$ but not differentiable on $(0, \infty)$
 (c) there exists $\alpha > 1$ such that $|f'(x)| < |f(x)|$ for all $x \in (\alpha, \infty)$
 (d) there exists $\beta > 0$ such that $|f(x)| + |f'(x)| \leq \beta$ for all $x \in (\alpha, \infty)$

SECTION III

Paragraph Type

Paragraph for Question Nos. 20 to 21

The circle $x^2 + y^2 - 8x = 0$ and hyperbola intersect at the points A and B.

- Q 20. The equation of a common tangent with positive slope to the circle as well as to the hyperbola is
- (a) $2x - \sqrt{5}y - 20 = 0$ (b) $2x - \sqrt{5}y + 4 = 0$ (c) $3x - 4y + 8 = 0$ (d) $4x - 3y + 4 = 0$
- Q 21. The equation of the circle with AB as its diameter is
- (a) $x^2 + y^2 - 12x + 24 = 0$ (b) $x^2 + y^2 + 12x + 24 = 0$
 (c) $x^2 + y^2 + 24x - 12 = 0$ (d) $x^2 + y^2 - 24x - 12 = 0$

Paragraph for Question Nos. 22 to 24

Consider the polynomial $f(x) = 1 + 2x + 3x^2 + 4x^3$. Let s be the sum of all distinct real roots of $f(x) = 0$ and let $|s| = t$.

- Q 22. The real number s lies in the interval
- (a) $\left(-\frac{1}{4}, 0\right)$ (b) $\left(-11, -\frac{3}{4}\right)$ (c) $\left(-\frac{3}{4}, -\frac{1}{2}\right)$ (d) $\left(0, \frac{1}{4}\right)$
- Q 23. The area bounded by the curve $y = f(x)$ and the lines $x = 0$, $y = 0$ and $x = t$, lies in the interval
- (a) $\left(\frac{3}{4}, 3\right)$ (b) $\left(\frac{21}{64}, \frac{11}{16}\right)$ (c) $(9, 10)$ (d) $\left(0, \frac{21}{64}\right)$
- Q 24. The function $f'(x)$ is
- (a) increasing in $\left(-t, -\frac{1}{4}\right)$ and decreasing in $\left(-\frac{1}{4}, t\right)$
 (b) decreasing in $\left(-t, -\frac{1}{4}\right)$ and increasing in $\left(-\frac{1}{4}, t\right)$
 (c) increasing in $(-t, t)$
 (d) decreasing in $(-t, t)$

Paragraph for Question Nos. 25 to 27

Tangents are drawn from the point $P(3, 4)$ to the ellipse $\frac{x^2}{9} + \frac{y^2}{4} = 1$ touching the ellipse at points A and B.

- Q 25. The coordinates of A and B are
- (a) $(3, 0)$ and $(0, 2)$ (b) $\left(-\frac{8}{5}, \frac{2\sqrt{161}}{15}\right)$ and $\left(-\frac{9}{5}, \frac{8}{5}\right)$
 (c) $\left(-\frac{8}{5}, \frac{2\sqrt{161}}{15}\right)$ and $(0, 2)$ (d) $(3, 0)$ and $\left(-\frac{9}{5}, \frac{8}{5}\right)$
- Q 26. The orthocentre of the triangle PAB is
- (a) $\left(5, \frac{8}{7}\right)$ (b) $\left(\frac{7}{5}, \frac{25}{8}\right)$ (c) $\left(\frac{11}{5}, \frac{8}{5}\right)$ (d) $\left(\frac{8}{25}, \frac{7}{5}\right)$

- Q 27. The equation of the locus of the point whose distances from the point P and the line AB are equal, is
- (a) $9x^2 + y^2 - 6xy - 54x - 62y + 241 = 0$ (b) $x^2 + 9y^2 + 6xy - 54x + 62y - 241 = 0$
- (c) $9x^2 + 9y^2 - 6xy - 54x - 62y - 241 = 0$ (d) $x^2 + y^2 - 2xy + 27x + 31y - 120 = 0$

SECTION IV

Matrix-Match Type

- Q 28. Match the statements in **Column I** with those in **Column II**.
[Note: Here z takes values in the complex plane and $\text{Im}(z)$ and $\text{Re}(z)$ denote the imaginary part and real part of z respectively.]

Column I

(A) The set of points z satisfying

$$|z - i| |z| = |z + i| |z|$$

is contained in or equal to

(B) The set of points z satisfying

$$|z + 4| + |z - 4| = 10$$

is contained in or equal to

(C) If $|\omega| = 2$ then the set of points

$$z = \omega - 1/\omega$$

is contained in or equal to

(D) If $|\omega| = 1$ then the set of

$$z = \omega + 1/\omega$$

is contained in or equal to

Column II

(p) an ellipse with eccentricity $\frac{4}{5}$

(q) the set of points z satisfying

$$\text{Im}(z) = 0$$

(r) the set of points z satisfying

$$|\text{Im}(z)| < 1$$

(s) the set of points z satisfying

$$|\text{Re}(z)| \leq 2$$

(t) the set of points z satisfying

$$|z| \leq 3$$

- Q 29. Match the statements in **Column I** with the values in **Column II**.

Column I

(A) A line from the origin meets the lines

$$\frac{x-2}{1} = \frac{y-1}{-2} = \frac{z+1}{1} \text{ and}$$

$$\frac{x-8/3}{2} = \frac{y+3}{-1} = \frac{z-1}{1} \text{ at P and Q}$$

respectively. If the length PQ = d the d^2 is

(B) The values of x satisfying

$$\tan^{-1}(x+3) - \tan^{-1}(x-3) = \sin^{-1}(3/5) \text{ are}$$

(C) Nonzero vectors $\vec{a}, \vec{b}$ and $\vec{c}$ satisfy

$$\vec{a} \cdot \vec{b} = 0, (\vec{b} - \vec{a}) \cdot (\vec{b} + \vec{c}) = 0, 2|\vec{b} + \vec{c}| = |\vec{b} - \vec{a}|.$$

If $\vec{a} = \mu \vec{b} + 4 \vec{c}$ then the possible values of μ are

(D) Let f be the function on $[-\pi, \pi]$ given by

$$f(0) = 9 \text{ and } f(x) = \frac{\sin 9/2x}{\sin 1/2} \text{ for } x \neq 0.$$

Then value of $2\pi \int_{-\pi}^{\pi} f(x) dx$ is

Column II

(p) -4

(q) 0

(s) 5

(t) 6

SECTION V

Integer Type

- Q 30. Let $\omega = \cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3}$. Then the number of distinct complex number z satisfying $\begin{vmatrix} z+1 & \omega & \omega^2 \\ \omega & z+\omega^2 & 1 \\ \omega^2 & 1 & z+\omega \end{vmatrix} = 0$ is equal to
- Q 31. Let $S_k, k = 1, 2, 3, \dots, 100$ denote the sum of the infinite geometric series whose first term is $\frac{k-1}{k}$ and the common ratio is $\frac{1}{k}$. Then the value of $\frac{100^2}{100} + \sum_{k=2}^{100} |(k^2 - 3k + 1)s_k|$ is
- Q 32. The number of all possible values of θ , where $0 < \theta < \pi$, for which the system of equations $(y+z) \cos 3\theta = (xyz) \sin 3\theta$
 $x \sin 3\theta = \frac{2 \cos 3\theta}{y} + \frac{2 \sin 3\theta}{z}$
 $(xyz) \sin 3\theta = (y+2z) \cos 3\theta + y \sin 3\theta$
have a solution (x_0, y_0, z_0) with $y_0 z_0 \neq 0$, is
- Q 33. The number of values of θ in the interval $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ such that $\theta \neq \frac{1}{5}n\pi$, where $n = 0, \pm 1, \pm 2$ and $\tan \theta = \cot 5\theta$ as well as $\sin 2\theta = \cos 4\theta$ is
- Q 34. The maximum value of the expression $\frac{1}{\sin^2 \theta + 3 \sin \theta \cos \theta + 5 \cos^2 \theta}$ is
- Q 35. The line $2x + y = 1$ is tangent to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$. If this line passes through the point of intersection of the nearest directrix and the x-axis then the eccentricity of the hyperbola is
- Q 36. If $\vec{a}$ and $\vec{b}$ are vectors in space given by $\vec{a} = \frac{i-j}{\sqrt{5}}$ and $\vec{b} = \frac{2i+j+3k}{\sqrt{14}}$ then the value of $(2\vec{a} + \vec{b}) \cdot \{(\vec{a} \times \vec{b}) \times (\vec{a} - 2\vec{b})\}$ is
- Q 37. If the distance between the plane $Ax - 2y + z = d$ and the plane containing the lines $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4}$ and $\frac{x-2}{3} = \frac{y-3}{4} = \frac{z-4}{5}$ is $\sqrt{6}$ then $|d|$ is
- Q 38. Let f be a real-valued differentiable function on $\mathbb{R}$, the set of real numbers, such that $f(1) = 1$. If the y-intercept of the tangent at any point $P(x, y)$ on the curve $y = f(x)$ is equal to the cube of the abscissa of P then the value of $f(-3)$ is equal to
- Q 39. For any real number x , let $[x]$ = largest integer less than or equal to x . Let f be a real valued function defined on the interval $[-10, 10]$ by is odd
- $$f(x) = \begin{cases} x - [x] & \text{if } [x] \\ 1 + [x] - x & \text{if } [x] \end{cases}$$
- Then the value of $\frac{\pi^2}{10} \int_{-10}^{10} f(x) \cos \pi x \, dx$ is
- Q 40. Let $a_1, a_2, a_3, \dots, a_{11}$ be real numbers satisfying $a_1 = 15, 27 - 2a_2 > 0$ and $a_k = 2a_{k-1} - a_{k-2}$, for $k = 3, 4, \dots, 11$. If $\frac{a_1^2 + a_2^2 + \dots + a_{11}^2}{11} = 90$ then the value of $\frac{a_1 + a_2 + \dots + a_{11}}{11}$ is equal to
- Q 41. Two parallel chords of a circle of radius 2 are at a distance $\sqrt{3} + 1$ apart. If the chords subtend at the centre, angles of magnitude $\frac{\pi}{k}$ and $\frac{2\pi}{k}$ where $k > 0$ then the value of $[k]$, the largest integer less than or equal to k , is
- Q 42. Consider a $\triangle ABC$. Suppose $BC = 6, CA = 10$ and the area of the triangle is $15\sqrt{3}$. If $\angle ACB > 90^\circ$ and r denotes the radius of the incircle of the triangle then r^2 is equal to

Q 43. Let k be a positive real number and

$$A = \begin{bmatrix} 2k-1 & 2\sqrt{k} & 2\sqrt{k} \\ 2\sqrt{k} & 1 & -2k \\ -2\sqrt{k} & 2k & -1 \end{bmatrix} \text{ and } B = A^{-1} = \begin{bmatrix} 0 & 2k-1 & \sqrt{k} \\ 1-\sqrt{k} & 0 & 2\sqrt{k} \\ -\sqrt{k} & 2\sqrt{k} & 0 \end{bmatrix}.$$

If $\det(\text{adj } A) + \det(\text{adj } B) = 10^6$ then $[k]$ is equal to

[Note: $\text{Adj } M$ denotes the adjoint of the matrix M and $[k]$ denotes the largest integer $\leq k$.]

Q 44. Let f be a function defined on $\mathbb{R}$, the set of real numbers, such that

$$f'(x) = 2010(x-2009)(x-2010)^2(x-2011)^3(x-2012)^4 \text{ for all } x \in \mathbb{R}.$$

If g is a function defined on $\mathbb{R}$ with values in $(0, \infty)$ such that

$$f(x) = \log_e g(x) \text{ for all } x \in \mathbb{R}$$

then the number of points in $\mathbb{R}$ at which g has a local maximum is

Answers

1b 2a 3d 4a 5d 6a 7b 8c 9a 10d

11b 12b 13c 14c 15a,c,d 16b 17c,d 18a 19b,c 20b

21a 22c 23a 24b 25d 26c 27a

28. p q r s t

A	☐	☒	☒	☐	☐
B	☒	☐	☐	☐	☐
C	☒	☐	☐	☒	☒
D	☐	☒	☒	☒	☒

29. p q r s t

A	☐	☐	☐	☐	☒
B	☒	☐	☒	☐	☐
C	☐	☒	☐	☐	☐
D	☐	☐	☒	☐	☐

30. 30 31 32 33 34 35 36

0	0	0	0	0	0	0
1	1	1	1	1	1	1
2	2	2	2	2	2	2
3	3	3	3	3	3	3
4	4	4	4	4	4	4
5	5	5	5	5	5	5
6	6	6	6	6	6	6
7	7	7	7	7	7	7
8	8	8	8	8	8	8
9	9	9	9	9	9	9

31. 37 38 39 40 41 42 43 44

0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1
2	2	2	2	2	2	2	2
3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4
5	5	5	5	5	5	5	5
6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9